

1. QTc prolongation is caused by which electrolyte disturbance? **Hypocalcaemia**. Rest of options were hyper

➤ **Risk factors for prolonged QTc include:**

- Congenital long Q-T syndrome,
- Underlying heart disease, bradycardia, heart failure, and ischemic disease
- Female gender,
- Extremes of age.
- Presence of liver disease,
- Electrolyte abnormalities (**hypokalaemia, hypocalcaemia and hypomagnesaemia**),
- Illicit drug use (principally stimulants).
- Starvation or anorexia.
- High physical exertion (agitation),
- High dosages of the drug contributing to the lengthened Q-T interval, and
- Rapid infusion of teratogenic drugs

2. Jittery crying baby, overarousal, cause? Maternal heroin consumption/ could be cocaine too.

☞ **Jittery → think Opioids or Cocaine**

☞ **If overarousal, tachycardia and tachypnoea → Cocaine withdrawal**

☞ **High-pitching crying, irritability, tremors → Opioids withdrawal**

[SPMM] Sarah Hughes is a 25-year-old woman with a 2-hour-old newborn who is noted to be very irritable, restless, tremulous and inconsolably crying. Which of the following substances did her mother most likely use during pregnancy?

Select one:

A. Heroin

B. Nicotine

C. Cocaine

D. Cannabis

E. Alcohol

Heroin withdrawal tends to be seen within 24 hours of birth, as with other short acting opioids. Signs and symptoms include excessive crying, tremors, increased muscle tone and reflexes. Similar withdrawal presentations are seen with other drugs, although this is less frequent with alcohol and cannabis.

3. Microcephaly, growth retardation, was born 2 kg- **Foetal alcohol syndrome**

Drugs in perinatal psychiatry

Which of the drugs provided in the options menu, when taken by a mother while pregnant, is most likely to be implicated in each case scenario?

1. 1 year old with cleft palate and delayed milestones ✕

2. A child with microcephaly, challenging behaviour, and cognitive deficits who weighed 2kg at birth ✕

3. Jittery newborn with **overarousal** and tachypnoea 2 hours after delivery ✕

There is strong evidence that valproate is associated with increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language, lower intellectual ability), and autism.

Clinical features of **fetal alcohol spectrum disorder** include microcephaly, abnormal facial features, growth deficits, mild intellectual disability, and associated behavioural problems (such as hyperactivity and sleep problems).

Stimulant use (such as cocaine) prior to delivery is associated with a neonatal withdrawal syndrome including symptoms of agitation, vomiting, and tachypnoea.

4. Safest antidepressant for breastfeeding? sertraline . other options escitalopram

Breast feeding & psychotropic medication (2)	
Choose the most appropriate drug for each of the following situations;	
A 33-year-old woman delivered a female baby 4 months ago and is breast-feeding. She has now been diagnosed with bipolar disorder. You have decided to start her on a mood stabilizer. Which one should be avoided in this case?	Choose... - Sodium valproate - Amitriptyline - Paroxetine - Venlafaxine - Fluoxetine - Desipramine - Doxepin - Carbamazepine - Dosulepin - Lithium - Citalopram
A 29-year-old mother who is currently nursing her 6-month-old child has now been diagnosed with depression. You are keen to start her on SSRIs for her depression	
A 27-year-old breast-feeding mother has been diagnosed with depression. She has had depression in the past and responded to one of the older antidepressants. She insists trying one of the older drugs again.	

Explanation: Lithium is contraindicated during breast-feeding (Kohen, 2005). Serum concentration and blood count should be monitored for infants exposed to lithium. 40- 50% of maternal serum level is excreted in breast milk. Infant serum level can rise up to 200% of maternal serum concentration (range 5%-200%). Serum level of lithium can be raised due to diminished renal clearance in neonates.

SSRI and tricyclic antidepressant (except Doxepin) are generally safe. **Paroxetine has a lower milk/plasma ratio than fluoxetine and sertraline.** Fluoxetine and its active metabolite norfluoxetine were detected in plasma and breast milk but not in infant's plasma.

The preferred tricyclic antidepressants for use in nursing mothers are Amitriptyline and Imipramine. Clomipramine and nortriptyline also do not show any adverse effects in infants. Maternal plasma and breast milk concentration of tricyclic antidepressant is the same. N-desmethyldoxepine, longer acting metabolite of doxepin, may accumulate in infants and cause severe drowsiness and respiratory depression

The correct answer is: A 33-year-old woman delivered a female baby 4 months ago and is breast-feeding. She has now been diagnosed with bipolar disorder. You have decided to start her on a mood stabilizer. Which one should be avoided in this case? → **Lithium**, A 29-year-old mother who is currently nursing her 6-month-old child has now been diagnosed with depression. You are keen to start her on SSRIs for her depression → **Paroxetine**, A 27-year-old breast-feeding mother has been diagnosed with depression. She has had depression in the past and responded to one of the older antidepressants. She insists trying one of the older drugs again. → **Amitriptyline**

5. Question about prevalence of types of eating disorders. Which one is correct?

- A. Bulimia is more common than anorexia.
- B. anorexia and bulimia are equally common.
- C. increased variation in different developed countries

[SPMM] With respect to the epidemiology of bulimia and anorexia, choose the incorrect statement.
Select one:

- A. The prevalence of bulimia is around 2%
- B. The stability of diagnoses of eating disorders is poor
- C. The prevalence of anorexia is around 0.5% to 1%
- D. Anorexia is excessively represented in lower social classes**
- E. Bulimia generally starts later than anorexia

Explanation: Though it was once thought that anorexia nervosa (AN) was more common in higher social classes, this is not reflected in the current literature. AN can affect individuals of all social classes. The increase of AN in low-income and middle-income countries suggests that cultural traditions associated with industrialisation, urbanisation, and globalisation might be associated with environmental risk constellations for the development of AN. It is possible that western influences could increase the number of individuals who engage in behaviours such as strict dieting or excessive exercise, which can then trigger eating disorders in genetically susceptible individuals.

Epidemiology of Eating Disorders (ref NICE Guidelines 2017, Morris and Anderson 2020)
Prevalence of anorexia nervosa (AN) in Western countries: 1% females and < 0.5% males
Sex ratio in adults is 1:8 (men : women); less skewed in young people
AN is rare in children <13 years and relatively rare among middle-aged and elderly women
Peak age of onset of anorexia nervosa: 15-19 years
Anorexia nervosa has highest rate of mortality among all mental disorders; most common causes of death are suicide (20%) and cardiac complications
Bulimia nervosa (BN) is more prevalent than AN, with estimates of 1-4% in young women and lower in men; age of onset of BN is usually later than AN
Age at identification of BN is decreasing, currently at 15-24 years old
About half of those with BN have a history of AN, especially binge eating-purging type
Lifetime prevalence of binge-eating disorder (BED): approx 2% for women and 0.3% for men
Compared with other eating disorders, BED is more common in males and older people
BED is commonly associated with obesity and increased risk of mortality



6. History of sexual assault and learning disability. These are static risk factors, dynamic, depends on the gender

Risk factors for any untoward incident (suicide, crime or violence) can be categorized as:

- **Static risk factors:** These are fixed and historical: e.g. family history of suicide. They cannot be modified.
- **Stable risk factors:** These are long term, enduring issues but are modifiable to some extent and not fixed: e.g. diagnosis of personality disorder.

- **Dynamic risk factors** :fluctuate markedly in both duration and intensity: the e.g. presence of acute anxiety symptoms or akathisia. A dynamic risk factor can act synergistically and multiply the effect of underlying static and stable risk factors if not addressed promptly.

Certain dynamic factors may occur only in future (e.g. upon discharge, a patient may feel helpless).

[SPMM] Mr Green is a 35 year old man who is due to be released from prison. Whilst an inmate, serving a sentence for grievous bodily harm and theft he was treated in the healthcare wing for depression and was also diagnosed with antisocial personality disorder. Which of the following factors would be associated with an increased risk of re-offending?

Select one:

- A. Returning to his parent's home immediately following release
- B. Having an index offence which was violent in nature
- C. Suffering from depression
- D. Returning to amphetamine use following release**
- E. Having served a 10 year sentence

Dynamic factors, which are associated with reoffending, include drug misuse, alcohol misuse, impulsivity, relationship with anti-social peers, lack of family relationships, lack of employment, having unsuitable accommodation. Static factors which are associated with a reduced risk of reoffending include: first time in custody, employment in the 12 months before custody, being worried about having spent time in prison, being older, longer sentences (greater than 1 year). Therefore returning to amphetamine use following discharge would be associated with an increased risk of reoffending.

[SPMM] Mr Daniel Alwin is a 28-year-old man who has been diagnosed with dissocial personality disorder. He has recently been released from prison after serving a sentence for actual bodily harm. He was reviewed by a psychologist prior to release to determine his risk to others in the future. Which of the following is most accurate regarding short to medium term risk assessment?

Select one:

- A. Risk factors for violent and sexual offending are the same
- B. Static risk factors are most important to consider
- C. The relevant risk factors for male and female offenders are the same
- D. Substance misuse history is unlikely to modify risk
- E. Dynamic risk factors are most important to consider**

Dynamic risk factors, such as substance misuse, engagement with relevant interventions and expression of anger, are thought to be more relevant for risk assessment in the short term. Static risks, such as childhood experiences, previous violent incidents and demographic factors, are thought to be more relevant over the longer term.

[SPMM] The percentage of new offenders among those charged of indecent exposure is

Select one:

- A. 50%
- B. 10%
- C. 40%
- D. 80%**
- E. 5%

With indecent exposures, the offender usually stops reoffending after a conviction; serious progressive crimes are very rare. Nearly 20-30% reoffend and if such reoffence leads to a conviction, the prognosis is very poor in terms of high recidivism.

[SPMM] What percentage of people convicted of indecent exposure have no previous record of sexual offences?

Select one:

- A. 20%**
- B. 50%
- C. 80%
- D. 5%
- E. 10%

20% are first-time offenders and have no previous record of sexual offences. 60% of them have a previous history of sexual offences. 70% of them have a history of previous sexual and non-sexual offences. Nearly 20-30% of them reoffend.

7. New-born born with cleft palate, has cardiac malformation, developmental delay- maternal valproate

See Q. 2

8. Persistent anxiety and social anxiety is most linked to - schizotypal personality disorder, schizoid, ocd

9. Chronic use of ketamine causes?

- A. Bladder atrophy,**
- B. bleeding,
- C. renal failure,
- D. liver failure

[SPMM] A man who has been abusing multiple substances now complains of frequent urination and passing blood in the urine. The most likely offending drug is

Select one:

- A. Cannabis
- B. Heroin
- C. Crack
- D. LSD
- E. Ketamine**

Explanation: Ketamine is structurally similar to phencyclidine and is used as an anaesthetic. It is a unique anaesthetic, as it does not depress the reticular activating system (thus does not usually knock out consciousness completely) but reduces the cortical awareness of painful stimuli. It is taken illicitly as a sniffed powder. The use of ketamine can cause urinary tract symptoms (e.g. **frequency, urgency, urge incontinence, dysuria, and haematuria**). The causal agent has not been determined, but direct irritation by ketamine and/or its metabolites is a possibility. Investigations have revealed interstitial cystitis, detrusor overactivity, and decreased bladder capacity. Symptoms generally settle several weeks after stopping ketamine. **Up to 30% of Ketamine users have urinary symptoms.**

10. A new test is compared with another one to test validity- concurrent validity

Validity and Reliability	
Match the following scenarios to their description of validity or reliability:	
1. Multisource feedback in for doctors in Foundation Year 2 (F2) was in line with multisource feedback in Foundation Year 1 (F1)	✗
2. Two respondents on multisource feedback gave similar positive reports on the same F1 trainee.	✗
3. Two radiologists reported the same degree of hippocampal atrophy and this was associated with a diagnosis of Alzheimer's dementia	✗
4. A new screening questionnaire for depression was being evaluated and compared to the gold standard that was administered at the same time to see if it could still diagnose depression with fewer items or questions.	✗

Inter-rater reliability
 Test re-test reliability
 Concurrent validity
 Construct validity
 Content validity
 Predictive validity

Explanation: Inter-rater reliability is used to assess the degree to which different raters/observers give consistent estimates of the same phenomenon. Having good test re-test reliability signifies the internal validity of a test and ensures that the measurements obtained in one sitting are both representative and stable over time. Concurrent validity is a measure of how well a particular test correlates with a previously validated measure. Predictive validity involves testing a group of subjects for a certain construct, and then comparing them with results obtained at some point in the future. Construct validity refers to whether a scale or test measures the construct adequately. It is a broader concept and is best opted for if other options are not specific enough.

1. Multisource feedback in for doctors in Foundation Year 2 (F2) was in line with multisource feedback in Foundation Year 1 (F1) – **Predictive validity**
2. Two respondents on multisource feedback gave similar positive reports on the same F1 trainee. - **Inter-rater reliability**
3. Two radiologists reported the same degree of hippocampal atrophy and this was associated with a diagnosis of Alzheimer's dementia - **Inter-rater reliability**
4. A new screening questionnaire for depression was being evaluated and compared to the gold standard that was administered at the same time to see if it could still diagnose depression with fewer items or questions. - **Concurrent validity**

11. Alcohol withdrawal, patient is in acute liver failure what to use? **Oxazepam**

[SPMM] John Masters is a 45-year-old man with cirrhosis secondary to hepatitis C and chronic alcohol dependence. He is admitted to hospital with confusion and is found to have decompensated liver disease and encephalopathy. He is agitated and is refusing medications. Which of the following medications would be appropriate to treat alcohol withdrawal in this gentleman?

Select one:

- A. Disulfiram
- B. Diazepam
- C. Chlordiazepoxide
- D. Oxazepam**
- E. Lorazepam

[SPMM] Usually chlordiazepoxide is the agent of choice for treating alcohol withdrawal. However, in patients with cirrhosis or advanced liver disease the shorter half-life of lorazepam and absence of active

metabolites with oxazepam may prevent prolonged effects and accumulation and make these better choices. Lorazepam is the correct response to this question as it is available IM if needed (the patient is refusing oral medication at this point). Diazepam is also available IM but has a long half-life. Disulfiram is a treatment for alcohol addiction but is not appropriate in the acute withdrawal state.

[SPMM] In patients with hepatic insufficiency, the best option available to prevent delirium tremens is
Select one:

- A. Oxazepam**
- B. Clonazepam
- C. Carbamazepine
- D. Chlordiazepoxide
- E. Clonidine

Oxazepam undergoes only phase 2 metabolism and does not rely on hepatic sufficiency for safe administration.

[SPMM] Mr. Y was assessed by the drug and alcohol team, who recommended alcohol detoxification. He is a chronic smoker and has a history of a chronic respiratory disease. Which one of the following is the most suitable drug for detoxification?

Select one:

- A. Chlordiazepoxide
- B. Clonazepam
- C. Oxazepam**
- D. Temazepam
- E. Diazepam

Explanation: In the setting of alcohol detoxification, shorter-acting benzodiazepines (e.g. oxazepam) may be preferred in those for whom over-sedation must be avoided, in people with liver disease who may not be able to metabolise long-acting agents efficiently, and in people with COPD. However, there is a greater risk of breakthrough seizures and greater potential for misuse.

12. Which one acts both on GABA a and NMDA:

- A. Alcohol**
- B. Ketamine**

Mechanism	Examples
Drugs which interfere with ionotropic receptors or ion channels	Alcohol, nicotine, benzodiazepines, ketamine
Drugs which interfere with G coupled receptors	Opioids, cannabinoids, y-hydroxybutyrate (GHB)
Drugs that target monoamine transporters	Amphetamine, ecstasy, cocaine

13. Most important (or common) defence mechanism according to ICD in Borderline?

- A. Idealisation/devaluation**
- B. Acting out

[SPMM] Klein's depressive position is related to the process of learning to cope with which of the following?

Select one:

- A. Depression
- B. Isolation
- C. Hunger
- D. Sexual needs
- E. Ambivalence**

Explanation: Melanie Klein viewed projection and introjection as the primary defensive operations in the first months of life. Infants project derivatives of the death instinct into the mother and then fear attack from the 'bad mother', a phenomenon that Klein referred to as persecutory anxiety. This anxiety is intimately associated with the paranoid-schizoid position, infants' mode of organising experience in which all aspects of infant and mother are split into good and bad elements. As the disparate views are integrated, infants become concerned that they may have harmed or destroyed the mother through the hostile and sadistic fantasies directed toward her. At this developmental point, children have arrived at the depressive position, in which the mother is viewed ambivalently as having both positive and negative aspects, and as the target of a mixture of loving and hateful feelings.

14. Alzheimer's most likely finding on CT - **hippocampal atrophy**

15. What is a must for causation according to Susser's criteria:

- A. Time order,**
- B. Specificity,
- C. Consistency,
- D. Predictive performance

[SPMM] Which of the following is an essential criterion of causation as per Susser's Criteria in Epidemiology?

Select one:

- A. Direction**
- B. Specificity
- C. Consistency
- D. Strength of association
- E. Predictive performance

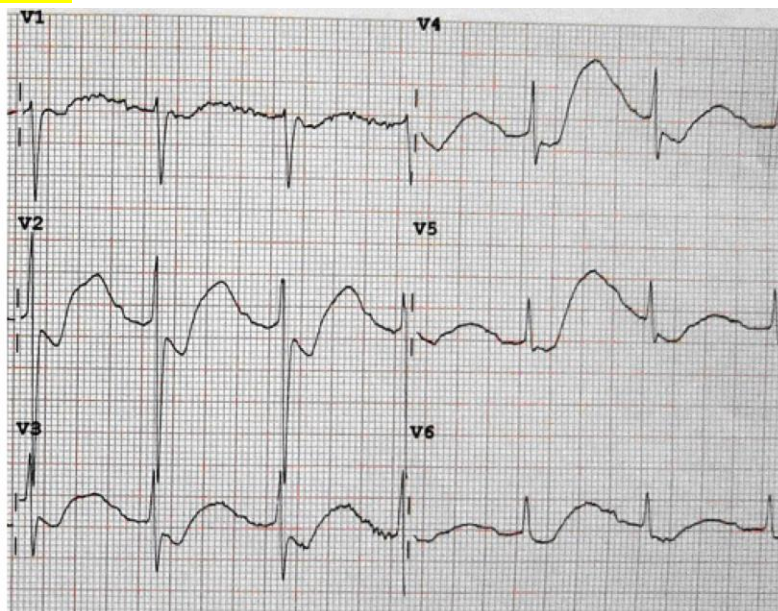
According to Mervyn Susser, to determine causality between a risk factor A and an effect B, 3 elements are essential—association, time order, and direction. He called this the "triad of essential causal properties". Of these, direction requires the ability to make a direct attribution between A and B without this being explained by confounders (thus weighs on the internal validity of research study seeking causality). Then, there are 5 additional elements—strength, specificity, consistency, predictive performance, and coherence – that are not essential but adds to the evidence.

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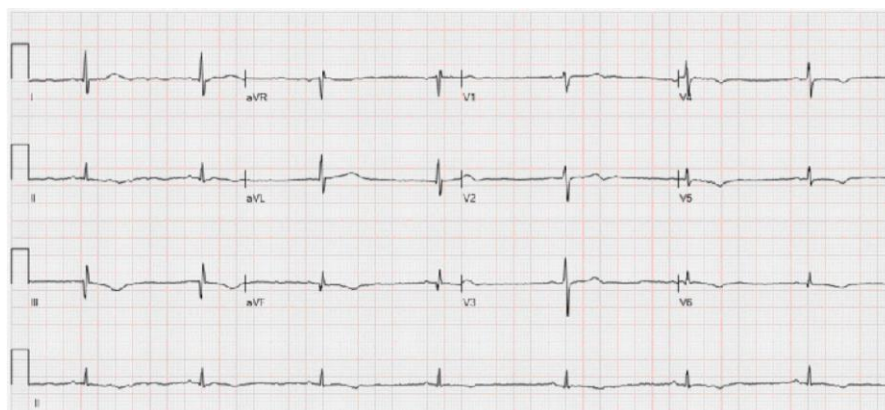
16. Clozapine, tachycardia, high troponin, what is the ECG finding? **Saddle type elevations**, t inversion
See Q.94 Fresh Q. 2022



17. Anorexic girl, has been abusing diuretics and laxatives, presents with lethargy and muscle cramps likely ecg finding (EMI q) hypokalaemia findings **ST segment depression with U waves**
See Q.94 Fresh Q. 2022



18. Donepezil increased to 10 mg yesterday, hypotensive, has ankle oedema what's ECG finding- **sinus bradycardia**



19. What favours schizophrenia diagnosis than autism diagnosis:

- A. Mood incongruent symptoms,
- B. Catatonia,
- C. Near normal development,**
- D. Ritualistic

Catatonia has been added as a modifier to ASD in DSM5

[SPMM] Please match the scenarios described with the most likely cause for catatonia.

A 42-year old man is changed from an oral antipsychotic, to a depot injection. A week after the depot he complains of stiffness in his arms, and the following day he lies rigidly in bed, sweating profusely, with his eyes moving around and not able to rest on your face as you attempt to talk to him

Choose...
Neuroleptic Malignant Syndrome
Depressive Stupor
Manic stupor
Autism
Schizophrenia
Postencephalitic state
Intracranial space-occupying lesion

A 17-year old young man has been brought onto the ward after an aggressive outburst at home. He is now standing very still, with his torso leaning forwards, and has kept this position for two hours. You try to engage him in conversation, and his father tells you that he does not communicate verbally, and never has done

A 22 -year old woman is brought to hospital by her friends after she stopped moving several hours previously. They tell you that for the past three days she has been out of the house all day and all night, that she reported feeling completely wonderful, and had spent all her savings on unusual jewellery

Explanation:

Q1: This man is now presenting with features of Neuroleptic malignant syndrome, which includes muscular rigidity, hyperthermia and diaphoresis. Administering a depot injection is considered as one of the risk factors for developing **NMS**.

Q2: A young man who does not have verbal speech and who has had an aggressive outburst may suggest he has Autism. People with **Autism** can sometimes enter catatonic states or maintain postures for long periods of time.

Q3: The history preceding the stupor is suggestive of **mania** (elated mood, excessive spending, lack of sleep/increased energy).

20. Victims of female homicide perpetrator mostly belong to the age group <16 (mock 3, q 82)

[SPMM] Victims of female homicide perpetrators with psychosis often belong to the age group of;

Select one:

- A. 60-75
- B. <16**
- C. 16-35
- D. >75
- E. 36-60

The victims of female homicide perpetrators are often in their own family. A substantial number of these victims are <16 years of age, especially if the perpetrator has psychosis. 50% of mentally ill perpetrators

of child homicide have a psychotic disorder. But note that fathers are significantly more likely to kill their children than mothers, if one considers all child homicides. Also 13% of perpetrators of filicide (child killing by parents) took their own life after killing their child (this is called filicide-suicide)

[SPMM] In females, which of the following age group is associated with the highest risk of committing a homicide?

Select one:

- A. 40-49
- B. <16
- C. 20-29
- D. 30-39**
- E. 16-19

Over 40 per cent of female perpetrators of intimate partner homicide are aged between 30 and 39' - the average age being 35 years

21. Types of costs- sick leave- indirect cost

[SPMM] In an attempt to study the cost of adherence vs. non-adherence to treatment in depression treated at primary care centres, an investigator calculates the costs of illness in terms of the costs for hospitalization, GP-visits, specialist referral, cost of medications, and cost of each day of sick leave due to depression. Which of the following among the above can be called an indirect cost?

Select one:

- A. Cost of specialist referral
- B. Cost of medications
- C. Cost of hospitalisation
- D. Cost of GP visits
- E. Cost of sick absence**

Direct costs: Cost of hospital space, cost of drugs, equipments, salary of doctors and other professionals etc. **Indirect costs:** Costs due to **loss of productivity** for the patient, cost of having a caretaker for one's dog or a nanny for one's child when hospitalized etc. Intangible (not measurable) costs: Pain and suffering, social stigma, physical discomfort, etc.

22. Cost of drug- **direct cost**

23. PCR : the four procedures of PCR pipette , electricity and research material etc were present however reader is a person , ~~seemed to be indirect costs~~ – **Direct Cost**

24. Pain and stigma- **intangible cost**

25. Describes picks disease, has no insight, decline in cognitive functions, what would be the type of dysphasia this person develops? **Primary motor dysphasia**

Note: As frontotemporal dementia progresses, the symptoms of the three clinical variants can converge, as an initially focal degeneration becomes more diffuse and spreads to affect large regions in the frontal and temporal lobes.

Type	Key issue	% of cases	Mean age onset	Characteristic symptoms	Earliest atrophic areas
bvFTD	Behaviour	54-69	58	Poor personal and social decorum Disinhibition Poor judgement and problem solving Apathy (often mistaken for depression) Compulsive / perseverative behaviour Hyperorality or dietary changes Loss of empathy	Anterior cingulate Frontotemporal (R > L)
nfvPPA Non-Fluent	Speech production	14-35	63	Slow / slurred speech Decreased word output and phrase length Word finding difficulties Apraxia of speech Spared single word comprehension (can understand single words)	Dominant frontal operculum Premotor area Supplementary motor area Anterodorsal insula
svPPA Semantic variant	Semantic understanding	9-12	59	Intact speech fluency Word finding difficulties (anomia) Impaired single word comprehension (can't) Repetitive speech Reduced word comprehension	Left anterior temporal Amygdala

26. Asks for calculation of positive likelihood ratio from given sensitivity and specificity at a certain cut off point answer is 4

- **LR+** = sensitivity / 1- specificity
- **LR-** = 1- sensitivity / specificity

27. Therapies for autistic children. A kid has visual program to plan his activities, what can you use to show him afternoon program? **Now and then cards**

Interventions in patients with intellectual disabilities
Which type of therapy is being utilised in the clinical scenarios below, as applied to patients with intellectual disabilities?

1. A patient with limited ability to use verbal communication reacts positively when others mirror his body language and pleasantly imitate his vocalisations. ✖

2. A nurse attempts to initiate changes in a patient's daily routine by providing him with a visual display. ✖

3. A support team wants to understand why a patient becomes agitated whenever his relatives visit him and develop a plan to better support him and his family. ✖

Intensive interaction
Sensory processing
Makaton therapy
Behavioural activation
Cognitive behavioural therapy
Positive behavioural support
Now and next cards

Explanation:

Intensive interaction is a 'body-language' based approach used to develop the communication and social inclusion of those who are preverbal, such as people diagnosed with intellectual disabilities and/or autism. The approach consists of using pre-verbal social communication techniques, such as mirroring gestures, communicative moves, and vocalisations.

Now and next cards display planned activities in symbols (words, pictures, photographs, icons, actual objects) that are understood in the order in which they will occur. They present the abstract concepts of activities and time in concrete forms by using pictures, words, and other visual elements to describe what will happen, in what order, and where.

Positive behavioural support is a multicomponent framework for developing an understanding of the challenging behaviour displayed by an individual and using this understanding to develop, implement, and evaluate the effectiveness of a personalised and ensuring system of support.

28. He is scared by his mother visiting him, what can you do about that? Social stories

[SPMM] A mother comes to visit her son with autism, and he becomes agitated. What technique should be used to help him understand the visits?

Select one:

- A. **Social stories**
- B. Intensive interaction
- C. Makaton therapy
- D. Now and next cards
- E. Cognitive behavioural therapy

Carol Gray's social stories are short, highly individualised stories that provide children with social information. They help children with autism spectrum conditions more easily understand a situation personal to them such as learning life skills, new experiences, understanding emotions (or people), transitions, or change. The child is included in the narrative, often with a range of helpful information.

29. Imitate LD child's body signs and language: **Intensive interaction techniques**

30. What's the gene involved in early onset Alzheimer's dementia of downs syndrome:

- A. **Amyloid**
- B. Apoe4.
- C. Apoe2.
- D. Apoe1.
- E. Other option tau

[SPMM] Of the following, the risk of developing Alzheimer disease in Down syndrome is most increased by:

Select one:

- A. Obesity
- B. Chronic anxiety
- C. A family history of Alzheimer disease
- D. Mild intellectual disability rather than severe intellectual disability
- E. Eczema

Explanation: Almost all adults with Down syndrome (DS) develop Alzheimer disease (AD)-like neuropathology by 40 years of age. AD is characterised by a long asymptomatic preclinical stage in which amyloid pathology develops approximately 15-20 years before any cognitive impairment is observed. This preclinical stage is present in the autosomal dominant forms of AD and in early-onset AD associated with DS. The evidence is compelling that an increased dosage of **APP (an HSA1 gene that encodes amyloid precursor protein) in trisomy 21 is necessary** for an increased risk of AD in people with DS. The cognitive decline needs to be understood in the context of the existing cognitive

phenotype and individual risk, including the APOE genotype, as the presence of the APOE E4 allele seems to increase risk of earlier cognitive decline due to AD.

The genetics of Alzheimer's disease

- * Presenilin 2 gene (Chromosome 1) – early onset
 - * Presenilin 1 gene (Chromosome 14) – early onset
 - * Beta Amyloid precursor protein gene (Chromosome 21)- the gene for Amyloid precursor protein is found on the long arm and is implicated in the early onset dementia. The gene for amyloid precursor protein is on the long arm of chromosome 21. The beta amyloid protein, the major constituent of senile plaques, is a 42-amino acid peptide that is a breakdown product of amyloid precursor protein. People with Down syndrome (trisomy 21) have three copies of the amyloid precursor protein gene. Another cause of excessive deposition of the beta amyloid protein is a mutation on codon 717 in the Amyloid precursor protein gene. Amyloid PET scanning is a
- incidence of Alzheimer's disease than no $\epsilon 4$ gene, and people with two $\epsilon 4$ genes have the disease eight times more frequently than do those with no $\epsilon 4$ gene. Genetic testing for presence of $\epsilon 4$ is not currently recommended as it is also present in people without dementia, and not found in all cases of dementia

31. First trimester, mixed affective state, what to use. Options: Olanzapine was not in option

- A. Lithium,
- B. Quetiapine,**
- C. Valproate,
- D. An uncommon antipsychotic

[SPMM] A patient with bipolar disorder and a history of mixed affective states is in the first trimester of her pregnancy. She has not been on any medications of late, but is very concerned about the possibility of relapse. Of the following, what would be the preferred medication to prescribe to her?
Select one:

- A. Olanzapine**
- B. Valproate and lithium
- C. Lithium
- D. Valproate
- E. Lamotrigine

Explanation: For pregnant women with bipolar disorder who have had a long period without relapse, the possibility of switching to a safer drug (antipsychotic) or withdrawing treatment completely before conception and for at least the first trimester should be considered. No mood stabiliser is clearly safe. NICE recommends the use of mood stabilising antipsychotics as a preferable alternative to a mood stabiliser. **ECT can be considered in pregnancy**

- ➔ **Quetiapine** is safe in pregnancy as It has relatively low rate of crossing placenta
- ➔ **Olanzapine, quetiapine and risperidone** are safe in breastfeeding

32. Tardive dyskinesia on flupentixol depot and procyclidine 20 mg. Patient has been stable on the current dose of flupentixol and procyclidine for 20 years. What do you do first?

- A. Lower the dose of depot,
- B. Increase procyclidine,
- C. Lower procyclidine,**
- D. Add Tetrabenazine,
- E. Switch to atypical antipsychotic

	Dystonia	Pseudo-parkinsonism (e.g. tremor)	Akathisia	Tardive dyskinesia
Prevalence (with older drugs)	Approximately 10% But more common in - young males - neuroleptic-naïve - high potency drugs (e.g. haloperidol)	Approximately 20% But more common in - elderly females - those with pre-existing neuro damage (e.g. stroke)	Approximately 25%	5% of patients per year of antipsychotic exposure But more common in - elderly women - those with affective illness - those who have had EPSE early on in treatment
Time taken to develop	Acute dystonia can develop within minutes or hours of starting antipsychotics	Days to weeks after antipsychotic started or dose increased	Hours to weeks	Months to years
Treatment	- Anticholinergic drugs* - Switching to alternative antipsychotic - Botulinum toxin	- Reduce the dose - Anticholinergic drugs* - Switching to alternative antipsychotic	- Reduce the dose - Switching to alternative antipsychotic (quetiapine/ olanzapine) Propranolol Mirtazapine or mianserin (low dose) - Anticholinergic drugs* - Cyproheptadine - Benzodiazepine (e.g. diazepam or clonazepam) - Clonidine	Stop anticholinergic (if prescribed) Reduce the dose Switch to an atypical (clozapine and quetiapine have best evidence) Tetrabenazine Ginkgo biloba

33. Anorexia patient, what electrolyte disturbance is the most worrying? Potassium (other options are bicarbonate but no phosphate)

Eating disorders
Identify the metabolic abnormalities commonly seen in the following clinical situations

1. A 16-year-old girl with history of significant weight loss, body image distortions and morbid fear of fatness have some endocrine abnormalities in her blood ✖

2. A 23-year-old woman with history of bulimia nervosa and is determined to reduce weight by inducing repetitive vomiting ✖

3. A 20-year-old woman uses regular laxatives for last 2 years to reduce weight has some electrolyte abnormalities (Choose TWO) ✖ ✖

4. A 16-year-old girl was admitted to the psychiatric unit for re-feeding as her weight was dangerously low. She died suddenly during re-feeding (choose TWO) ✖ ✖

5. A 15-year-old girl with a history of anorexia was admitted to the medical ward for severe dehydration. During fluid resuscitation, she went into a state of coma. ✖

Hypernatraemia
Hyponatraemia
Hyperkalemia
Hypokalemia
Hypocortisolemia
Hyperparathyroidism
Hyperphosphatemia
Hypophosphatemia
High T3 and high growth hormone
Low T3 and high growth hormone

Explanation: The diagnosis is anorexia nervosa. The common endocrine changes seen are low concentrations of luteinising hormone, follicle stimulating hormone, oestradiol, low T3, T4 in low normal range, normal concentrations of thyroid stimulating hormone (**low T3 syndrome**, Mild increase in plasma cortisol and raised growth hormone concentration.

Electrolyte disturbances are common in those who frequently vomit or misuse large quantities of laxatives or diuretics. In repetitive vomiting, loss of hydrochloric acid from gastric juices leads to **metabolic alkalosis** (loss of acid - alkalosis) along with **hypokalemia**.

Laxative misuse can lead to metabolic acidosis, **hyponatremia**, **hypokalemia**. A large amount of bicarbonate may be lost in the stool in laxative induced diarrhea. With normal functioning kidneys, the lost bicarbonate is replaced effectively, and, therefore, a serious base deficit does not develop. When there is poor renal blood flow due to hypovolemia or starvation, base deficit, and acidosis develop rapidly. Acidosis also results from excessive production of lactic acid when patients have severe diarrhoea.

Refeeding syndrome usually occurs within four days of starting to feed. Patients can develop fluid and electrolyte disorders, especially **hypophosphatemia** along with neurologic, pulmonary, cardiac, neuromuscular, and hematologic complications. Most effects result from a sudden shift from fat to carbohydrate metabolism and a sudden increase in insulin levels after refeeding which leads to increased cellular uptake of phosphate. This syndrome can occur at the beginning of treatment for anorexia nervosa when patients are reintroduced to a healthy diet. It can lead to acute heart failure. If hypovolemia is rapidly corrected, it can cause **hyponatremia** leading to central pontine myelinolysis and coma

Correct Answer is : 1. Low T3 and high growth hormone 2. Hypokalemia 3. Hyponatremia, Hypokalemia 4. Hypokalemia, Hypophosphatemia 5. Hyponatremia

	Red: High impending risk to life	Amber: Alert to high concern for impending risk to life	Green: Low impending risk to life		Red: High impending risk to life	Amber: Alert to high concern for impending risk to life	Green: Low impending risk to life
Medical history and examination				ECG abnormalities			
Weight loss	Recent loss of weight of >1kg/week for 2 weeks (consecutive) in an undernourished patient ²⁶ Rapid weight loss at any weight, e.g. in obesity or ARIID	Recent loss of weight of 500-99g/week for 2 consecutive weeks in an undernourished patient ²⁶	Recent weight loss of <500g/week or fluctuating weight		• <18 years: QTc >460ms (female), >450ms (male) • 18+ years: QTc >450ms (females), >430ms (males) • And any other significant ECG abnormality	• <18 years: QTc >460ms (female), >450ms (male) • 18+ years: QTc >450ms (females), >430ms (males) • And no other ECG anomaly • Taking medication known to prolong QTc interval	• <18 years: QTc <460ms (female), <450ms (male) • 18+ years: QTc <450ms (females), <430ms (males)
BMI and weight	• Under 18 years: mSBMI ²⁷ <70% • Over 18: BMI <13	• Under 18: mSBMI 70-80% • Over 18: BMI 13-14.9	• Under 18: mSBMI >80% • Over 18: BMI >15				
HR (awake)	<40	40-50	>50				
Cardio-vascular health²⁸	Standing systolic BP below 64th centile for age or less than 90 if 18+, associated with recurrent syncope and postural drop in systolic BP of >20mmHg or increase in HR of over 30bpm (35bpm in <18 years)	Standing systolic BP <64th centile or <90 if 18+, associated with occasional syncope; postural drop in systolic BP of >15mmHg or increase in HR of up to 30bpm (35bpm in <18 years)	• Normal standing systolic BP for age and gender with reference to centile charts • Normal orthostatic cardiovascular changes • Normal heart rhythm	Biochemical abnormalities²⁹	• Hypophosphataemia and falling phosphate • Hypokalaemia (<2.5mmol/L) • Hypoalbuminaemia (<3mmol/L) • Hyponatraemia • Hypocalcaemia • Transaminases >3x normal range • Inpatients with diabetes mellitus: HbA1c >10% (86mmol/mol)		
Assessment of hydration status	• Fluid refusal • Severe dehydration (10%): reduced urine output, dry mouth, postural BP drop (see above), decreased skin turgor, sunken eyes, tachypnoea, tachycardia	• Severe fluid restriction • Moderate dehydration (5-10%): reduced urine output, dry mouth, postural BP drop (see above), normal skin turgor, some tachypnoea, some tachycardia, peripheral oedema	• Minimal fluid restriction • No more than mild dehydration (<5%): may have dry mouth or concerns about risk of dehydration with negative fluid balance	Haematology	• Low white cell count • Haemoglobin <10g/L		
Temperature	<35.5°C tympanic or 35.0°C axillary	<36°C	>36°C	Disordered eating behaviours	Acute food refusal or estimated calorie intake <500kcal/day for 2+ days		
Muscular function³⁰ SUSS Test	Unable to sit up from lying flat, or to get up from squat, at all or only by using upper limbs to help (Score 0 or 1)	Unable to sit up or stand from squat without noticeable difficulty (Score 2)	Able to sit up from lying flat and stand from squat with no difficulty (Score 3)				
Muscular function: Hand grip strength³¹	Male <30.5kg Female <17.5kg (3rd percentile)	Male <38kg Female <23kg (5th percentile)	Male >38kg Female >23kg				
Muscular function: MUAC³²	<18cm (approx. BMI<13)	18-20cm (approx. BMI=15.5)	>20cm (approx. BMI >15.5)				
Other clinical state	Life-threatening medical condition, e.g. severe haematemesis, acute confusion, severe cognitive slowing, diabetic ketoacidosis, upper gastrointestinal perforation, significant alcohol consumption	Non-life-threatening physical compromise, e.g. mild haematemesis, pressure sores	Evidence of physical compromise, e.g. poor cognitive flexibility, poor concentration				

34. What antidepressant for 8 year old depressed patient? **Fluoxetine**

NICE Guidelines limit their recommendation to young people between the age of 5-18.
The following table summarises their recommendations.

Severity of depression	Recommendation
Mild depression (including dysthymia) - without significant comorbid problems or active suicidal ideas or plans	Watchful waiting (no intervention and review in 2 weeks) Digital CBT Group CBT Group IPT Group non-directive supportive therapy (NDST) (If not suitable then individual CBT or attachment-based family therapy)
Moderate to severe depression (continued mild depression despite 2-3 months of psychology is treated as moderate / severe)	5-11 year olds: Family based IPT Family therapy (family-focused treatment for childhood depression and systems integrative family therapy) Psychodynamic psychotherapy Individual CBT +/- fluoxetine 12-18 year olds: Individual CBT +/- fluoxetine (If not suitable then IPT for adolescents, family therapy (attachment-based or systemic), brief psychosocial intervention or psychodynamic psychotherapy)
Treatment resistant depression or depression with psychotic symptoms	Intensive psychological therapy +/- fluoxetine, sertraline, citalopram, augmentation with an antipsychotic

35. ECT cognitive impairment what's correct?

A. Lithium increases risk of cognitive impairment following ECT

B. cognitive impairment due to ECT is more common in males

C. unilateral ECT causes more cognitive impairment

[SPMM] Which one of the following best reflects the evidence on the adverse effects of electroconvulsive therapy (ECT) when used in the treatment of moderate to severe depression?

Select one:

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- A. Simulated ECT causes the same degree of memory loss as ECT immediately after treatment
- B. Bilateral ECT is more likely to cause cognitive impairment than unilateral ECT**
- C. ECT is more likely to cause profound cognitive impairment six months after treatment than simulated ECT
- D. Low dose ECT is more likely to cause transient memory loss than high dose ECT
- E. ECT causes no direct effects on memory in non-depressed individuals

Explanation:

The Royal College of Psychiatrists' statement on ECT (2017) notes the following:

- Cognitive effects are the main limitation to the wider use of ECT, particularly the occasional acute confusion shortly after the treatment, retrograde amnesia, and some losses in autobiographical (personal) memory longer term
- Bilateral ECT is more effective than unilateral ECT but may cause more cognitive impairment
- Right unilateral electrode placement, compared with the traditional bi-temporal placement, may minimise episodic and autobiographical memory deficits
- With unilateral ECT, a higher stimulus dose is associated with greater efficacy, but also increased cognitive impairment compared with a lower stimulus dose
- Continuation ECT is not associated with adverse memory outcomes

N.B: ECT + lithium is associated with 12-fold higher odds of delirium compared to ECT alone.

<https://www.sciencedirect.com/science/article/pii/S1935861X19303584#:~:text=ECT%20%2B%20lithium%20is%20associated%20with,lithium%20compared%20to%20ECT%20only>

36. What's element of motivational interviewing?

- A. Cognitive dissonance amplification,**
- B. Incentive,
- C. Goal setting

DEARS

D- Developing discrepancy

E- Empathy

A- Avoid arguments

R- Rolling with resistance

S- Support self- efficacy

37. Short, has freckles, lisch nodules, ld- **neurofibromatosis**

Clinical genetics

Choose the most appropriate option for each of the following descriptions.

This occurs almost exclusively in females and the associated gene is MECP2 at Xq28.

The penetrance is greater in males than females. The gene sequence has been cloned and designated as FMR-1.

A congenital disorder caused by a microdeletion of chromosome 22, associated with intellectual disability and schizophrenia.

An autosomal dominant condition with NFI gene 17q11.2 involved. The gene product is thought to be a tumour suppressor gene.

An X-linked recessive condition due to a mutation in the HGPRT gene.

Choose...

Prader-Willi syndrome
Williams syndrome
Angelman syndrome
Rett syndrome
Neurofibromatosis
Velo-cardio-facial syndrome
Down syndrome
Turner syndrome
Lesch-Nyhan syndrome
Fragile X syndrome

Choose...

Your answer is incorrect.

Explanation:

Rett syndrome is a degenerative disorder that almost exclusively affects females. It is believed to be caused by a dominant X-linked gene. Mutations in the MECP2 gene on the X chromosome are present in the majority of girls with Rett syndrome.

Fragile X syndrome is a common inherited cause of intellectual disability (ID) with X-linked dominant transmission. Penetrance is low, but greater in males than females due to the 'protective' effects of the second normal X chromosome in females. The syndrome is associated with a large sequence of triplet repeats (CGG) at a fragile site on the X chromosome (Xq27.3). The gene sequence is designated FMR-1.

Di George (velo-cardio-facial) syndrome is caused by a microdeletion of chromosome 22 (22q11.2). Approximately 50% of individuals have intellectual disability. Associated behavioural and psychiatric disorders include schizophrenia-like and blunted/inappropriate affect.

Also called von Recklinghausen disease, neurofibromatosis is the most common of the neurocutaneous syndromes caused by a single dominant gene, which may be inherited or occur as a new mutation. Neurofibromatosis type 1 is an autosomal dominant condition, with the responsible gene on chromosome 17 (approximately 50% spontaneous mutations).

Lesch-Nyhan syndrome is an extremely rare recessive condition due to a mutation in the HGPRT gene on the short arm of the X chromosome. Deficiency of the enzyme hypoxanthine phosphoribosyl transferase leads to build of uric acid in the body fluids (partial HGPRT deficiency results in gout).

38. Most suitable antipsychotic for egfr <25 ?

- A. Olanzapine,
- B. Amisulpride,
- C. Sulpiride

[SPMM] An 80-year-old woman with an eGFR 25 is to be prescribed an antipsychotic. Which of the following drugs would be the most appropriate for her?

Select one:

- A. Olanzapine**
- B. Chlorpromazine
- C. Clozapine
- D. Amisulpride
- E. Sulpiride

In the setting of renal impairment, no antipsychotic is clearly preferred to another. However, Sulpiride and Amisulpride are extensively renally cleared and should be avoided. Avoid highly anticholinergic agents (such as clozapine and chlorpromazine) because they can contribute to urinary retention. **Recommended choices are haloperidol and olanzapine (5 mg in general adults, 2.5mg in elderly).**

Drug group	Recommended psychotropics in renal impairment
Antipsychotics	No agent clearly preferred to another, however: Avoid sulpiride and amisulpride Avoid highly anticholinergic drugs because they can contribute to urinary retention First-generation antipsychotic: suggest haloperidol 2-6mg/day Second-generation antipsychotic: suggest olanzapine 5mg/day
Antidepressants	No agent clearly preferred to another, however: Citalopram (care re: QTc-prolonging effects) and sertraline are reasonable choices CBT where available
Mood stabilisers	No agent clearly preferred to another, however: Avoid lithium if possible Suggest starting valproate or lamotrigine at low dose, increase slowly, and monitor for adverse effects
Anxiolytics and hypnotics	No agent clearly preferred to another, however: Excessive sedation more likely to occur, so monitor all patients carefully Lorazepam and zopiclone are reasonable choices
Anti-dementia drugs	No agent clearly preferred to another, however: Rivastigmine is a reasonable choice

Hepatic impairment

Identify the most appropriate and/or effective drug/drugs to be prescribed for the following clinical situations

1. Antipsychotics recommended to treat psychosis in a patient with hepatic impairment ✗
2. Antidepressants recommended to treat depression in a patient with hepatic impairment ✗
3. Mood stabilizer of choice in hepatic impairment ✗
4. Mood stabilizers contraindicated in severe liver disease (Choose TWO) ✗
5. Sedative of choice in severe liver disease ✗

Chlorpromazine
Haloperidol
Trifluopromazine
Imipramine
Reboxetine
Sertraline
Sodium valproate
Gabapentin
Lithium
Oxazepam
Diazepam

Haloperidol (low dose) is considered as the drug of choice with sulpiride/amisulpride considered as second choice options.

Among antidepressants, there is best evidence for use of **Imipramine**. Start with 25 mg/day and titrate slowly if required. Among SSRIs, Paroxetine and citalopram are recommended. Avoid sertraline.

Lithium is the mood stabilizer of choice in hepatic impairment, with gabapentin as second choice though the efficacy of the latter is questionable.

Mood stabilizers contraindicated in severe liver disease include **sodium valproate and lamotrigine**.

Where necessary, use only small dose of short acting benzodiazepines like **lorazepam, oxazepam and temazepam**. Sedative drugs can precipitate hepatic encephalopathy.

Correct Answer is : 1. Haloperidol 2. Imipramine 3. Lithium 4. Sodium valproate, Gabapentin 5. Oxazepam

Table 8.7 gives an outline of recommended psychotropics in **hepatic impairment**.

Table 8.7 Psychotropics in **hepatic impairment**

Drug group	Recommended drugs
Antipsychotics	Sulpiride/amisulpride : no dosage reduction required if renal function is normal Paliperidone : if depot required
Antidepressants	Paroxetine, sertraline, citalopram, or vortioxetine : start at low dose. Titrate slowly (if required) as above.
Mood-stabilisers	Lithium : use plasma levels to guide dosage. Care needed if ascites status changes.
Sedatives	Lorazepam, oxazepam, temazepam : as short half-life with no active metabolites Use low doses with caution, as sedative drugs used in severe disease can precipitate hepatic encephalopathy Zopiclone : 3.75mg with care in moderate hepatic impairment

39. What's the most common early onset dementia? **Alzheimer's**

[SPMM] The 2nd most common degenerative disease before the age of 64 is

Select one:

- A. Alzheimer's dementia
- B. Normal pressure hydrocephalus

- C. Lewy body dementia
- D. Frontotemporal dementia
- E. Vascular dementia**

The 2nd common condition contributing to early onset dementia is vascular dementia, followed by frontotemporal dementia; **the most common condition is Alzheimer's disease**

40. Dementia scale for non-English speaker, with little education who has interpreter.

- A. RUDAS**
- B. MMSE
- C. MOCA
- D. ACE-III

41. Why do we calculate odds instead of RR in case control studies? **No new cases**

42. Visual hallucinations in a patient with Parkinson's. What medication? **Rivastigmine**

Parkinson's with Psychotic symptoms → Quetiapine / Clozapine

Parkinson's with psychotic symptoms and dementia → Rivastigmine

43. New born with cleft palate, cardiac malformation, developmental delay- **maternal valproate use. See Q. 2**

44. Strongest factor of repeating sexual offence:

- A. Sexual deviancy and antisocial,**
- B. Sexual assault with weapon,
- C. Homosexual with strangers,
- D. Child pornography with alcohol.
- E. Substance use

[SPMM] Which of the following factors is most strongly associated with sexual offence recidivism?
Select one:

- A. Mental illness
- B. Conditional release
- C. Part-time employment
- D. Deviant sexual interests**
- E. Lack of victim empathy

Explanation: Sexual offenders are more likely to recidivate (i.e., reoffend) with a non-sexual offence than a sexual offence. **10-20% of sexual offenders commit further sexual offences over 5-10 years.** Sexual recidivism has been shown to be associated with at least two factors: **deviant sexual interests and antisocial orientation/lifestyle instability.** Individuals with interests in deviant sexual activities (e.g., sexual interest in children, rape, auto-erotic asphyxia, and paraphilic disorders) are among the most likely to continue sexual offending. **Antisocial orientation** is also one of the largest predictors of sexual recidivism (e.g. non-compliance with supervision, violation of conditional release). Sexual crimes are associated with an unstable, antisocial lifestyle instability (crime-prone personality) characterised by rule violations, poor employment history and reckless, impulsive behaviour. Rapists are more likely to

have an antisocial orientation than are child molesters. Although general lack of empathy for others (a trait of psychopathy) is associated with recidivism, lack of empathy for one's victims specifically does not appear to be.

[SPMM] Mr Elsworth is a 39-year-old man who is an inmate in a category B prison following a sexual offence. He undergoes a risk assessment, accounting for historical and psychological factors that would influence the probability of him offending again. Which of the following factors would increase the risk of sexual reoffending the most?

Select one:

- A. **Dissocial personality disorder and Paedophilic paraphilia**
- B. Intra-familial incest
- C. Possession of child pornography
- D. History of depression
- E. Sexual assault with penetration of a stranger

Offenders who demonstrate a persistent lack of empathy and guilt, as in **dissocial personality disorder**, and who demonstrate evidence of sexual deviance, such as paedophilia, are most likely to sexually reoffend. **Those who sexually assault strangers, and particularly those who commit rape, are likely to reoffend as well**, but show more general criminal behaviour and will commit other violent offences.

45. Most likely to form physical dependence:

- A. **Alprazolam**,
- B. Methamphetamine

Benzodiazepine	Half life (hours)
Diazepam	20-100
Clonazepam	18-50
Chlordiazepoxide	5-30
Nitrazepam	15-38
Temazepam	8-22
Lorazepam	10-20
Alprazolam	10-15
Oxazepam	6-10
Zopiclone	5-6
Zolpidem	2
Zaleplon	2

46. Most likely/common effect of paroxetine use in pregnancy on new-born? **Irritability**, pulmonary hypertension (some say pulm hyp some say ECT)

[SPMM] A 29-year-old woman is 6 months pregnant. She is depressed, has a previous history of depression, and a family history of depression. She has responded well to paroxetine in the past. If she were to take paroxetine for the remainder of her pregnancy, which of the following effects is most likely to be seen in the newborn?

Select one:

A. Irritability

- B. Neural tube defects
- C. Facial abnormalities
- D. Ebstein anomaly
- E. Neonatal bleeding

Explanation: SSRIs do not appear to be major teratogens, with most data supporting the safety of fluoxetine. When taken late in pregnancy, SSRIs may increase the risk of persistent pulmonary hypertension of the newborn; the absolute risk is very low. Women who take SSRIs may be at increased risk of postpartum haemorrhage. The neonate may experience discontinuation symptoms, which are usually mild (such as agitation or irritability), or rarely respiratory distress and convulsions, with SSRIs. The risk is particularly high with short half-life drugs such as paroxetine and venlafaxine.

[SPMM] Which of the following would be the most appropriate medication to prescribe for a pregnant woman with first-episode depression?

Select one:

A. Sertraline

- B. Fluoxetine
- C. Paroxetine
- D. Citalopram
- E. Venlafaxine

Explanation: If initiating an antidepressant during pregnancy, previous response to treatment must be taken into account (i.e. consider using previously effective treatment). For previously untreated patients, consider sertraline (least placental exposure). When taken in late pregnancy, SSRIs may increase risk of persistent pulmonary hypertension the new-born, though the absolute risk is very low. The risk with sertraline may be lower than with other SSRIs. Neonates may experience discontinuation symptoms, usually mild, with SSRIs. The risk is particularly high with short half-life drugs such as paroxetine and venlafaxine



Recommendations – Depression in Pregnancy

(Adapted from Maudsley Prescribing Guidelines, 2021. pp. 687-689.)

- ☐ Patients who are already receiving antidepressants and are at **high risk of relapse** are best maintained on the same antidepressant during and after pregnancy.
- ☐ Those who develop a moderate-severe or severe depressive illness during pregnancy should be treated with antidepressant drugs.
- ☐ If initiating an antidepressant during pregnancy or for a woman considering pregnancy, previous response to treatment must be taken into account. The antidepressant that has previously been effective should be considered. For previously untreated patients, **sertraline** may be considered (**least placental exposure**).
- ☐ Screen for alcohol use.
- ☐ Be vigilant for the development of hypertension and preeclampsia.
- ☐ Women who take SSRIs may be at increased risk for postpartum haemorrhage.
- ☐ When taken late in pregnancy, SSRIs may increase the risk of persistent pulmonary hypertension of the newborn. The absolute risk is very low. The risk with **sertraline** may be lower than other SSRIs.
- ☐ The neonate may experience discontinuation symptoms, which are usually mild, such as agitation and irritability, or rarely respiratory distress and convulsions (with SSRIs). The risk may be particularly high with short half-life drugs such as **paroxetine and venlafaxine**. Continuing to breastfeed and then 'weaning' by switching to mixed (breast/bottle) feeding may help reduce the severity of reactions.

47. Diabetic retinopathy, diabetes type 2, not responded to aripiprazole what can you use?

- A. **Amisulpride**
- B. Haloperidol
- C. Risperidone

Summary: Antipsychotics – risk of diabetes and impaired glucose tolerance

High risk	Clozapine, olanzapine
Moderate risk	Quetiapine, risperidone, phenothiazines
Low risk	High-potency FGAs (e.g. haloperidol)
Minimal risk	Aripiprazole, amisulpride, asenapine, brexpiprazole, cariprazine, lumateperone, lurasidone, ziprasidone

48. Citalopram and diphenhydramine intoxication - **QTc prolongation** - **See Q.94 Fresh Q. 2022**

[SPMM] Medication is the most frequently identified contributing factor to acquired QTc prolongation. Both citalopram and diphenhydramine are associated with QTc prolongation

49. What is true about uncomplicated alcohol withdrawal: **starts in 4-12 hours after the last drink**

[SPMM] A 34-year-old patient who has been dependent on alcohol is involved in a road traffic accident and admitted to an orthopaedic ward for multiple fractures. Which of the following is correct with respect to his clinical state?
Select one:

Time course of alcohol withdrawal

- Onset of shakes (4-12 hours)
- Onset of perceptual disturbances (8-12 hours)
- Seizure onset (12-24 hours); peaks at 48 hours
- Delirium onset (72 hours)

A. Alcohol withdrawal symptoms usually start within 3-12 hours

- B. The chances of experiencing delirium is highest after 5 days
- C. He will not have a seizure unless he has a head injury
- D. Risk of seizures is maximal from day 3 to day 5
- E. Having withdrawal symptoms 5 days after abstinence is unusual

Explanation: Mild alcohol withdrawal symptoms usually start within 3-12 hours of the last drink and peak at 24-48 hours. They may last up to 14 days. Generalised seizures usually start within 12-18 hours of the last drink. They may commence before blood alcohol levels reach zero. Delirium tremens usually start 3-4 days after the last drink.

[Psych-Mentor] Note that withdrawal symptoms and seizures may occur before blood alcohol levels reach zero*

A small percentage of people (3-5%) undergoing withdrawal experience delirium tremens (DT). This usually occurs 3-4 days following alcohol cessation. The mortality rate for this condition is 10-20% if untreated*.

50. What's the cut-off age of ADHD in ICD11/DSM5: **12**

51. Dementia, hallucination, paranoia, whats first line- **Risperidone**

The following table summarises the main pharmacological options:

Drug class	Summary
Antipsychotics	NICE: Offer antipsychotic for aggression, agitation, distress, or psychosis only when other measures ineffective and there is severe distress or risk of harm to self / others Maudsley: Only use when need is balanced with increased risk of CVA Risperidone is first-line (as only licensed option). If can't use risperidone try alternative (olanzapine is a good option)
Cognitive enhancers	Worth a trial if antipsychotics not indicated / tolerated
Benzodiazepines	Avoid
Antidepressants	Some evidence for SSRI (sertraline and citalopram) for agitation in dementia Avoid tricyclics Use of SSRI is justified for depression in dementia
Mood stabilisers	Limited data to support use Valproate best avoided
Sedating antihistamines	Promethazine often used for sedation but may cause cognitive impairment - short-term use only
Melatonin	Limited evidence to support use, but safe to use and may be justified in some cases where benefits are seen

Dementia Management

Choose the single most appropriate treatment for each of the following clinical situations;

1. Harry Whitaker is an 80-year-old man with moderate Alzheimer's Dementia, who over the past few days has been more verbally hostile than usual to staff at his nursing home. ✖
2. Mrs May is 84 years old, has Alzheimer's Dementia, takes Memantine at an adequate dose, and is currently residing in a care home. She is in reasonable physical health. Mrs May has been physically aggressive to staff and visitors for several months, and everyone involved has worked hard to implement behavioural management techniques, with little success. ✖
3. Mr Singh is a 78-year-old man with Parkinson's Dementia whose hostility, agitation and aggression have been causing concern over the past few months. He has been investigated for physical health problems and nothing has been found to suggest he has an acute physical illness. ✖
4. Mr Churchill is a 79 year old man with Alzheimer' dementia. He developed dizziness and bradycardia with donepezil and Galantamine. The family would like him to take antidementia medications. ✖
5. Mrs Dorothy has Alzheimer's disease and she is on Antidementia medication for the last 2 years. She stopped eating as she thought her food was poisoned. She doesn't allow anyone in her family to enter inside her flat, as she believed that they are trying to harm her. ✖

Donepezil
Memantine
Ginkgo biloba
Haloperidol
Risperidone
Trazadone
Behavioural management
Rivastigmine
Clozapine.

Explanation:

1) **Behavioural management** should be tried as the first option in dementia patients presenting with challenging behaviour.

2) **Memantine** is as effective as Risperidone at alleviating difficulties relating to behavioural disturbances, but if Memantine has been trialled and not been effective, it is reasonable to trial Risperidone, taking into account potential adverse effects and associated risk factors.

3) **Rivastigmine** is associated with moderate improvement in dementia associated with Parkinson's Disease, although there are also unwanted effects (eg nausea, tremor).

4) **Memantine** can be used in patients who are first tried on acetyl cholinesterase inhibitors and developed side effects on taking them.

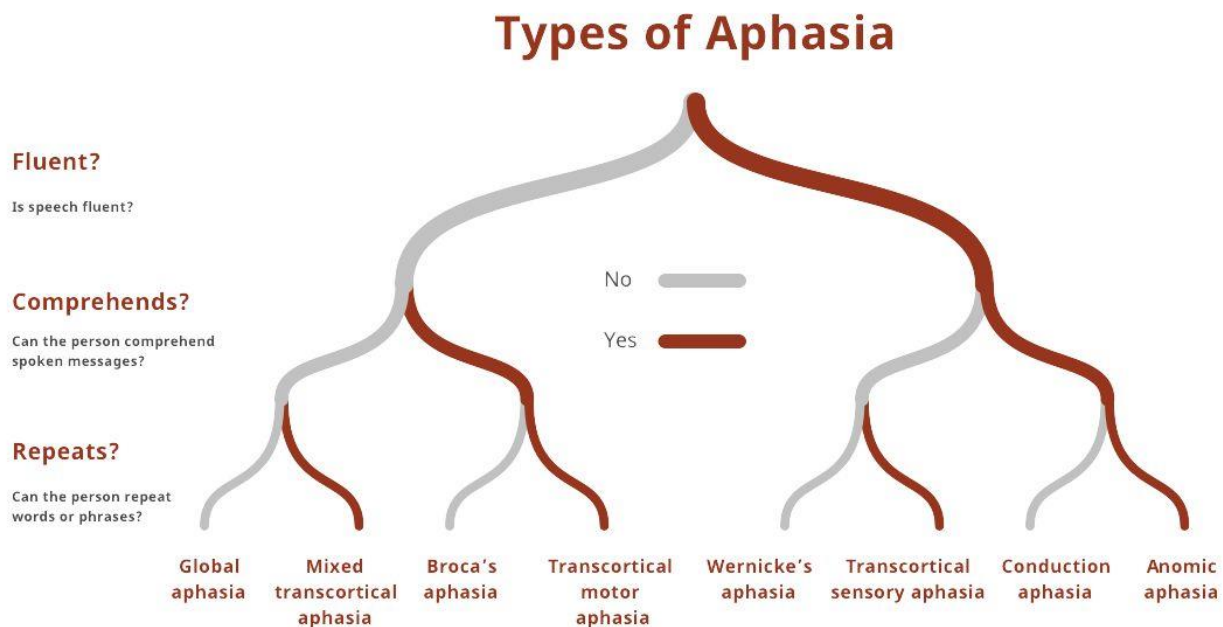
5) **Risperidone** should be used to treat her psychotic symptoms and it is licensed in the UK for the management of behavioural & psychological symptoms of dementia (BPSD).

52. Types of dysphasia, hears words knows that he hears a sound, can't understand what's being said :

Subcortical sensory dysphasia

53. Fluent speech, no syntax no grammar - **jargon dysphasia**

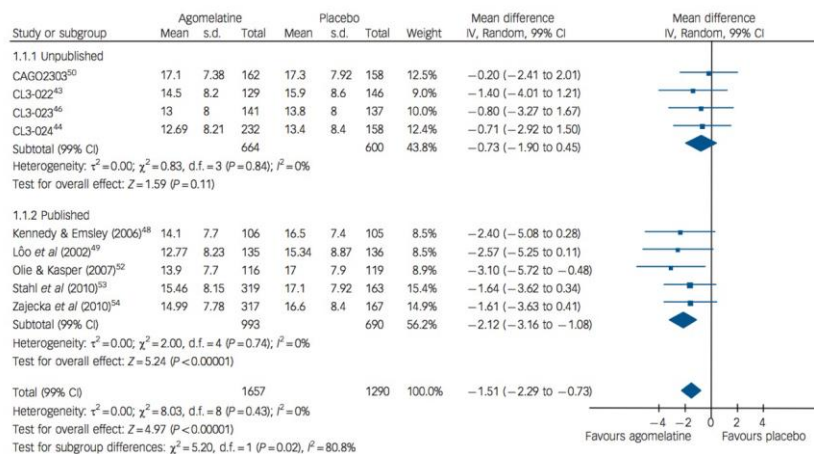
54. Understands everything, can speak, but cant repeat- **conduction aphasia**



www.aphasia.org

55. Forest plot showing meta analysis (agomelatine vs placebo study) what is the particular strength of this metanalysis? **Includes unpublished studies**

Mock 12 – Q. 93 – 97



56. According to the study, what can be said about agomelatine effectiveness for depression? **It is likely an effective antidepressant**

57. What do you use for literature search for the meta-analysis about agomelatine? Psychinfo google scholar, **Medline**

[SPMM] For healthcare interventions not restricted to a specific specialty, which database is commonly used to identify observational studies?
Select one:

- A. PsychINFO
- B. CINAHL
- C. MEDLINE**
- D. PsychLIT
- E. CENTRAL

For healthcare interventions, MEDLINE and Embase are the databases most commonly used to identify studies of all types. PsychINFO deals with psychology and other specialties allied to psychiatry. CINAHL covered nursing and allied mental health professions. CENTRAL (The Cochrane central register of controlled trials) has articles taken from a wide range of bibliographic databases and from other published and unpublished sources but covers controlled trials rather than observational studies.

Sources of Medical Evidence

Choose the most appropriate resource to access research literature from the below list:

To identify the recommended practice of occupational therapists when accepting referrals for PTSD

To identify guidelines on treatment of new onset adolescent depression

To identify the evidence reviewing a particular treatment option for cardiac failure in geriatric population

To identify the dominant psychoanalyst viewpoint of borderline personality disorder that existed in the 1870s in the United States of America

To identify a recent trial comparing the effectiveness of a new drug with an established drug

To locate a conference abstract reporting a highly controversial but novel findings on language development in schizophrenia

Choose...

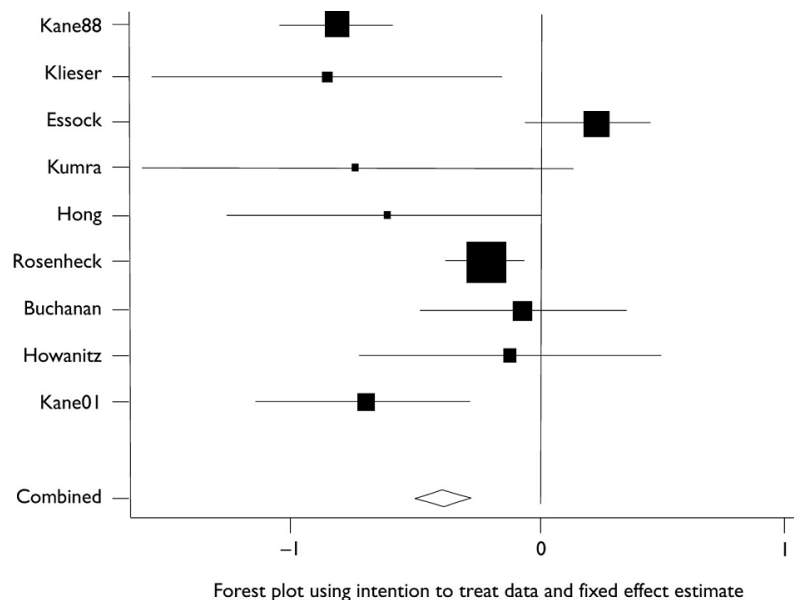
CINAHL
PubMed
NICE
PscINFO
Cochrane reviews
EMBASE
SIGLE

Choose...

The correct answer is: To identify the recommended practice of occupational therapists when accepting referrals for PTSD → CINAHL, To identify guidelines on treatment of new onset adolescent depression → NICE, To identify the evidence reviewing a particular treatment option for cardiac failure in geriatric population → **Cochrane reviews**, To identify the dominant psychoanalyst viewpoint of borderline personality disorder that existed in the 1870s in the United States of America → **PsycINFO**, To identify a recent trial comparing the effectiveness of a new drug with an established drug → **PubMed**, To locate a conference abstract reporting a highly controversial but novel findings on language development in schizophrenia → **SIGLE**

58. What's true about effect size in the forest plot?largest effect size with biggest square – **Rosenheck**

Mock 1 – Q. 107



59. Forest plot shown, How many studies show a statistically significant benefit of clozapine over conventional antipsychotics? **4**

60. Number of studies show statistically significant benefit of conventional antipsychotics over clozapine = **0**

Forest plot using intention to treat data and fixed effect estimate

Moncrieff, 2003 (Moncrieff, J. The British Journal of Psychiatry 2003; 183:161-166) reinvestigated the clozapine vs. conventional antipsychotics trials identified by a Cochrane review in a meta-regression analysis to study the sources of heterogeneity. The following forest plot was produced as a result. The pooled overall result favoured efficacy of clozapine. Choose the correct answers for each of the following questions from the names of trials listed above.

- The trial that is most likely to have the largest sample size among the above **x**
- The trial with most precise results **x**
- The trial that is most likely to have had longest follow up period **x**
- The trials whose outcomes did not cross the point of no difference for 95% confidence intervals (FOUR). **x**
 x **x** **x**

Correct Answer is : 1. Rosenheck 2. Rosenheck 3. Cannot be identified 4. Kane88, Klierer, Rosenheck, Kane01

61. 17 year old boy, acute confusion, bloods and CT normal. Not mentioning fever. what do you do first? **Urine drug screen**

[SPMM] A 20 year old young man presented with fever, acute onset of confusion and memory impairment. CSF shows normal glucose concentration, increased proteins and lymphocytosis. The most likely diagnosis is;
Select one:

- A. Lyme disease
- B. Herpes simplex encephalitis**
- C. Creutzfeldt Jakob disease
- D. Frederick's ataxia
- E. Motor neurone disease

HSE (Herpes simplex encephalitis) is the most common cause of limbic encephalitis. It is usually due to HSV-1. The clinical features include high fever, headaches, abrupt onset of confusion, change in personality and memory impairment, usually of acute onset. CSF findings include pleocytosis (raised white cell count) and elevated proteins. Viral encephalomyelitis leads to a lymphocytosis with normal CSF/plasma glucose ratio.

[SPMM] Mr Mathey is a 26-year-old man who has been brought into A&E with fever, confusion, agitation and auditory hallucinations over the last several hours. He has a lumbar puncture as part of his assessment, which demonstrates lymphocytosis, normal glucose and increased protein. Which of the following is the most likely diagnosis?
Select one:

- A. Herpes encephalitis**
- B. Lewy body dementia
- C. Huntington's disease
- D. CJD
- E. CADASIL

Herpes encephalitis is the most likely diagnosis in view of the acute onset of symptoms suggesting systemic inflammation and neurological involvement. Patients commonly have marked lymphocytosis, increased red cells, increased protein and low or normal glucose levels in CSF.

62. Shows a graph, what is it? **Scatter plot**

63. Scatter graph given, what does the line indicate? R is 0.7. Strong positive correlation, strong negative correlation, weak positive, weak negative
Q.2 correlation and regression

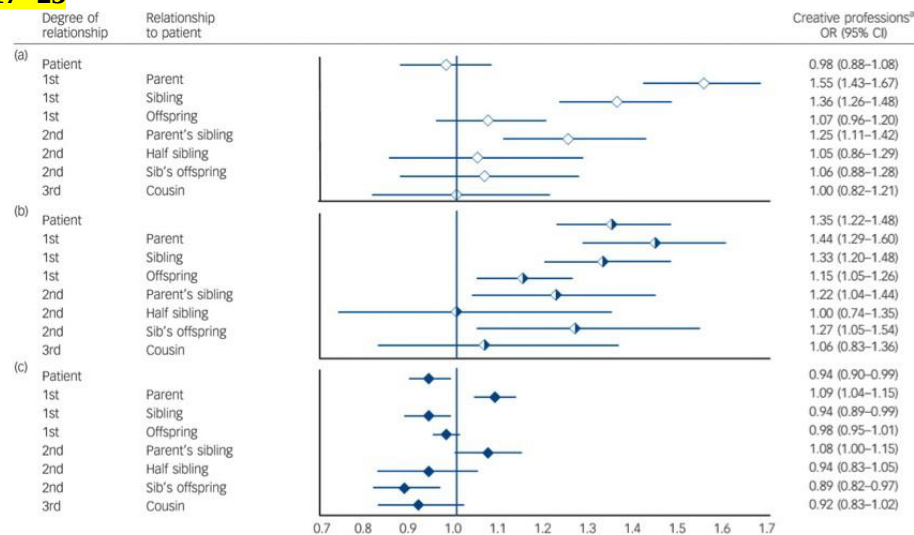
64. What test is used above? **Pearsons**

65. Forest plot is given. Which one is correct with regards to age? Older aged patients have more mental health disorder (p was <0.01 for age 65-70), there is no significant association of mental health disorders with age

66. Another forest plot.

67. Which relative is more likely to have a creative profession? Of schizophrenic patients? **Parents**

Mock 4, Q. 17- 23



68. Full agonist, causes paranoia, hallucinations **synthetic cannabinoid**

[SPMM] Anandamide receptors are associated with

Select one:

- A. Speed
- B. LSD
- C. Cocaine
- D. Heroin
- E. Cannabis**

Cannabis is the most commonly used illegal drug, with only a small minority of cannabis users ever using any other illegal drug. It is available in various forms that contain at least 60 psychoactive cannabinoids, the most important of which is 9-delta-tetrahydrocannabinol (THC). Cannabis usage can vary from infrequent situational use to heavy daily use. **There are specific cannabinoid receptors in the human brain, and the endogenous agonist at this receptor is called anandamide.** The role of the endocannabinoid system is yet to be defined. Anandamide (weak ligand) are formed from arachidonic acid - belong to endocannabinoid family. The two types of cannabinoid receptors, central (CB1) and peripheral (CB2), bind tetrahydrocannabinol (THC), the active ingredient of marijuana. Anandamide lowers intraocular pressure, decreases activity level, and relieves pain

69. Types of data: patients are divided according to age groups 35-50,50-60, etc- ordinal data

Data Types
For each of the following descriptions choose an appropriate response from given list:

A prospective study investigates relapse rates of depression in patients who have maintenance ECT.

A neuroimaging study measures insight using a single item as good, moderate, mildly abnormal, poor or absent to correlate with parietal volume.

Lean body mass on admission at an eating disorder unit.

Choose...

Ratio
 Multiple Nominal
 Ordinal
 Binary Nominal
 Interval
 Independent

The correct answer is: 1. Binary. 2. Ordinal. 3. Ratio

70. Whether or not risk assessment is done: **dichotomous data**

Data type	Description
Nominal	Observed values can be put into set categories which have no particular order or hierarchy. You can count but not order or measure nominal data (for example birthplace)
Ordinal	Observed values can be put into set categories which themselves can be ordered (for example NYHA classification of heart failure symptoms)
Discrete	Observed values are confined to a certain values, usually a finite number of whole numbers (for example the number of asthma exacerbations in a year)
Continuous	Data can take any value with certain range (for example weight)
Binomial	Data may take one of two values (for example gender)
Interval	A measurement where the difference between two values is meaningful, such that equal differences between values correspond to real differences between the quantities that the scale measures (for example temperature)
Ratio	A measurement where not only intervals but also ratios between numbers are meaningful due to a non-arbitrary zero point (for example weight, height)

71. Patient hospitalised 2 days ago, found hypotensive, not responding, bradycardic what to give:

- A. **Naloxone.**
- B. Naltrexone

72. What statistical test to compare categorical data? Compare ECT and the presence of stroke in 2 groups. Data is given. **Chi square**, Fisher's test, t test. *If data less than 5 could be Fisher's test*

73. What is dangerous driving most commonly linked to?

- A. **Dementia,**
- B. Mania
- C. Antisocial PD

[SPMM] The Driving Vehicle Licensing Agency (UK) requests that they are informed if people are diagnosed with certain mental illnesses as these may impact on their ability to drive. Which of the following diagnoses is most commonly associated with dangerous driving?

Select one:

- A. Depression
- B. Panic disorder
- C. Bipolar disorder
- D. Vascular dementia**
- E. Emotionally unstable personality disorder

Dementia, except potentially in the very early stages, is associated with significant risks around driving and the DVLA generally advises that people stop driving after they receive a diagnosis. The other disorders listed can be associated with dangerous driving but this is dependent upon the symptoms an individual experience, whether they are experiencing an episode of illness.

Dissocial personality disorder
Alcohol dependence
Schizophrenia
Asperger's syndrome
Depression
Intellectual disability
ADHD
OCD
Borderline personality disorder

Criminal behaviour:

Which of the following mental disorders would be most likely to be associated with the below scenarios?

1. Mr Young is a 22-year-old man who has been detained by the police for staring and waving at women in a park. When arrested he was surprised and did not appear to feel he was doing anything untoward. ✖

2. Mrs Wilkinson is a 45-year-old woman who is attending court after crashing into 3-parked vehicles whilst driving home. ✖

Explanation: 1) This case appears to be related to a poor understanding of societal norms. This form of behaviour can be seen in individuals with an **intellectual disability**. In such cases they tend to show no malice or attempt to avoid discovery or the police.

2) This would appear to represent **alcohol dependence or misuse**, as it appears the patient has crashed their car unintentionally whilst under the influence. One of the most common complications of alcoholism is road traffic accident.

<https://www.gov.uk/guidance/psychiatric-disorders-assessing-fitness-to-drive>

74. Sexual assault most common with?

- A. **Antisocial,**
- B. LD
- C. substance abuse

[SPMM] Which of the following offences is more common in people with intellectual disability compared to the general population?

Select one:

- A. **Sexual offences**
- B. Threats
- C. Substance misuse
- D. Theft
- E. Property damage

Explanation: Violent offences include common assault, murder, aggravated robbery, as well as any form of contact sexual offence. Sexual offences involve indecent assault, rape, and incest. Non-sexual non-violent offences include theft, property damage, substance misuse, threats, and arson. Compared with community groups, violent and sexual offending are more common in people with intellectual disability (ID). Among individual subject to pre-trial forensic psychiatric assessment, ID offenders are more likely than non-ID offenders to have a sexual crime as an index crime as well as previous convictions regarding sexual offending. These associations are restricted to male offenders; sexual offending is uncommon among women.

75. Theft like crimes-

- A. Antisocial,
- B. LD,
- C. Substance abuse

[SPMM] Which of the following disorders is least likely to be associated with shoplifting?

Select one:

A. Obsessive-compulsive disorder

- B. Schizophrenia
- C. Depression
- D. Substance misuse
- E. Personality disorders

Burglary, theft, and fraud are common offences that are rarely associated with psychiatric disorder. Shoplifting has attracted some clinical attention. About 5% of shoplifters have a significant mental disorder (personality disorder, substance misuse, depression, schizophrenia, dementia). Pure kleptomania (compulsive stealing) is extremely rare.

[Psych-Mentor] According to the National Treatment Outcomes Research Study (NTORS), what is the most common crime in those with drug dependence?

- A. Fraud
- B. Actual bodily harm
- C. Burglary
- D. Robbery

E. Shoplifting

Drug dependence is associated with a high incidence of criminal activity. Most of the offences are committed by a small proportion of people (10% of offenders account for 76% of the crimes). The National Treatment Outcomes Research Study (NTORS) found 61% of a drug misuse treatment sample reported committing crimes other than drug possession in the 3 months prior to starting treatment, with the most commonly reported offence being shoplifting.

76. Very noisy kid, severe LD, hypertelorism- **cri du chat**

77. MECP2 gene- **Retts syndrome**

Behavioural phenotypes in intellectual disability (1)
Choose one diagnosis for each of the behavioural phenotypes given below.

Stereotypic hand-wringing movements

Hyperactivity and cat-like cry during infancy

Self-injurious behaviour and twirling

Social disinhibition and excessive friendliness, pseudomature language abilities, and visuospatial motor deficits

Attraction to water and laughing at minimal provocation

Choose...

- Hurler syndrome
- Down syndrome
- Di George syndrome
- Williams syndrome
- San-Filippo syndrome
- Tuberous sclerosis
- Rett syndrome
- Cornelia de Lange syndrome
- Angelman syndrome
- Cri-du-chat syndrome

Explanation:

Rett syndrome is a degenerative disorder that almost exclusively affects females. There is initially normal development; then at ages 1-4 yrs, purposeful hand movements and spoken language are lost, and stereotypical hand movements (hand-wringing) and autism develop.

Features of **cri-du-chat syndrome** include a characteristic high-pitched cry that resembles a cat. Clinical features include microcephaly, rounded face, hypertelorism, micrognathia, dental malocclusion, epicanthic folds, low-set ears, hypotonia, and severe/profound intellectual disability.

Behavioural problems associated with **Cornelia de Lange** syndrome include self-injury, temper tantrums, stereotypic movements, and twirling.

Williams syndrome is associated with anxiety and fears. Often there is abnormal attachment behaviour manifested as anxiety, poor peer relationships, and hypersensitivity (or conversely as social disinhibition and excessive friendliness). Verbal skills are often better than motor and visuospatial skills.

Angelman ('happy puppet') syndrome is characterised by severe/profound intellectual disability. Approximately 86% of individuals have epilepsy. Other clinical features include happy disposition, paroxysmal laughter, absence of speech, hand flapping, clapping, ataxia (jerky limb movements, gait problems), and anecdotal love of water and music.

78. which test doesn't require normal distribution **mann whitney u**

79. DBT pre-treatment - motivation and agreements or priority action in DBT- self harm

DBT Developed as a treatment for Borderline Personality Disorder, especially to reduce self-harm. DBT addresses the difficulties faced by a patient with BPD in a hierarchical fashion starting from self-harming behaviours, moving on to therapy interfering behaviours and later behaviours reducing the quality of life.

80. Drug to lower ECT dose? Clozapine

Psychotropic class	Effect on seizure duration	Advice
Benzodiazepine	Reduced ↓	Avoid where possible
SSRIs	Minimal effect	
Venlafaxine	Minimal effect	
TCAs	Possibly increased ↑	TCAs are associated with arrhythmia following ECT in the elderly and those with cardiac disease so should be avoided in ECT in these groups
MAOIs	Minimal effect	
Lithium	Possibly increased ↑	Generally used in ECT without significant problems
Antipsychotics	Some potential increase in clozapine and phenothiazines, other antipsychotics considered ok ↑	Limited data
Anticonvulsants	Reduced ↓	If used as mood stabiliser continue but be prepared to use higher energy stimulus

81. Suicide risk in one year after self-harm 1/100

⇒ **0.7% (0.5 in females and 1.1 males)**

- ➔ Within 3 months of discharge from inpatient: 25 % and 40% before their scheduled follow up.
- ➔ Who visited GP in last one week: 40% (same 40% in older adults)
- ➔ More than 10 years after first attempt: 3%
- ➔ Life time prevalence of suicide : 4%
- ➔ Life time prevalence of suicidal thoughts: 15%

82. Most common comorbidity in OCD kids:

A. Depression.

B. GAD

Comorbidity: 70% have at least one co morbid disorder. OCD is commonly found to be comorbid with other psychiatric disorders, such as Tic disorders (17-40%), major depression (26%) specific developmental disabilities (24%), phobias and other anxiety disorders. There are also higher than expected rates of attention deficit/hyperactivity disorder (ADHD), autism spectrum disorders, and tic disorders, including Tourette's syndrome, among children and adolescents with OCD.

[SPMM] Which of the following is frequently co-morbid with generalised anxiety disorder (GAD) and shares substantial overlap in diagnostic criteria with GAD?

Select one:

A. Obsessive-compulsive disorder

B. Major depressive disorder

C. Schizophrenia

D. Bipolar disorder

E. Borderline personality disorder

Explanation: Generalized anxiety and depression are frequently comorbid: about 62% of patients with generalised anxiety disorder also have at least one episode of major depressive disorder during their lifetime. This finding is probably related to both a predisposition to mental illness and the substantial overlap in the diagnostic criteria for these disorders. Diminished ability to think or concentrate, restlessness, and sleep disturbance appear in the diagnostic criteria for both generalised anxiety and major depression.

83. Another question about OCD.

84. Which one has the highest comorbidity with OCD in adults?

A. OCPD,

B. GAD,

C. Major depression

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8631971/>

85. Definition given: Diagnostic overshadowing in LD

[SPMM] Diagnostic overshadowing is an often-debated issue in the psychiatry of learning disabilities. This concept refers to;

Select one:

- A. Making more than one diagnosis for a behavioural problem
- B. Providing an insufficient diagnostic explanation for genetically determined syndromes
- C. Making more than one diagnoses for a physical health problem
- D. Attributing a mental health issue to an already existing Learning disability**
- E. Attributing a psychiatric symptom to a behavioural phenotype

Diagnostic overshadowing is the term used which describes the tendency for clinicians to overlook symptoms of mental and physical health problems in clients/patients with learning disabilities and attribute them to being part of the "having a learning disability".

[SPMM] Miss Olive Charleston is a 32-year-old woman with moderate intellectual disability and has been suffering from otitis externa for the past 10 days. The staff members in her care home have observed her banging her head on doors and tables. They attributed this behaviour to her intellectual disability and have been trying to reassure and distract her without success. Which of the following best describes this phenomenon?

Select one:

- A. Retrospective diagnosis
- B. Institutionalization
- C. Diagnostic overshadowing**
- D. Stereotyping
- E. Neglect

Diagnostic overshadowing is the attribution of a person's symptoms to their mental condition, when such symptoms actually suggest a comorbid condition. The term was first used to describe the under-diagnosis of mental illness in people with intellectual disability. In recent years, the term has also been used when physical illnesses are overlooked in people with mental illness.

86. Intoxication q: laughing, giggling, relaxed. It is usually smoked- cannabis

Syndromes of substance use

Match each clinical presentation with the most appropriate intoxication or withdrawal syndrome from the options listed.

1. A 35-year-old man presents with high anxiety, nightmares, profuse sweating, and insatiable hunger. He admits to feeling 'low' recently and thinking about hanging himself. ✖
2. A 19-year-old woman describes feeling very well (despite her family's concerns that she seems unlike herself), is slurring her speech, and is somewhat unsteady. She can't stop giggling and repeatedly asks if she can have something to eat. ✖
3. A 55-year-old man is found on the street to be unresponsive and barely breathing. It is noted that his pupils are constricted. ✖
4. A 24-year-old woman presents with hallucinations and a feeling of being 'outside' herself, though she doesn't seem particularly concerned. On examination, she is noted to be tachycardic and hypertensive, with bidirectional nystagmus. ✖

Cocaine withdrawal is characterised by a 'crash' with symptoms of anxiety, tremulousness, dysphoric mood, lethargy, fatigue, nightmares (accompanied by rebound REM sleep), headache, profuse sweating, muscle cramps, stomach cramps, and insatiable hunger. The most serious withdrawal symptom is depression, which can be particularly severe and associated with suicidal ideation or behaviour.

Effects of **cannabis intoxication** include mild euphoria ('the giggles'), a sense of enhanced well-being, a subjective sense of enhanced sensation, relaxation, altered time sense, and increased appetite ('the munchies'). Physically, there is mild tachycardia and variable dysarthria and ataxia.

Acute medical problems associated with **heroin intoxication** include nausea and vomiting, constipation, respiratory depression, and loss of consciousness with aspiration (the cause of many fatalities). Physical effects also include pupillary constriction and changes in blood pressure, heart rate, and body temperature.

Ketamine can cause hallucinations and a dissociated state in which the patient has an altered sense of the body and reality and little concern for the environment. On physical examination, the patient may be hypertensive and tachycardic, and have increased salivation and bidirectional or rotatory nystagmus.

87. Short acting full agonist causing increased energy and decreased sleep- **Cocaine**

Physical signs/symptoms of intoxication

Five young men attend a substance misuse clinic with the following presentations. For each of the physical signs/symptoms of intoxication, select the most likely option from the list provided

Tachycardia, red eyes, lack coordination, postural hypotension	Choose... Cannabis intoxication Barbiturate intoxication Heroin intoxication Caffeine intoxication Nicotine intoxication Alcohol intoxication Benzodiazepines intoxication Amphetamine intoxication Cocaine intoxication LSD intoxication
Sedation, dizziness, respiratory depression	
Tachycardia, hypertension, anorexia, tremor and restlessness	
Tachycardia, tachypnea, hypertension, headache, chest pain and respiratory depression	
Pinpoint pupils, clammy skin, respiratory depression	

Explanation:

- 1) **Cannabis intoxication**- Tachycardia, red eyes, lack of coordination, postural hypotension.
- 2) **Benzodiazepines intoxication**- Sedation, dizziness, respiratory depression.
- 3) **Amphetamine intoxication**- Tachycardia, hypertension, anorexia, tremor and restlessness. (less respiratory symptoms as per Modyluy)
- 4) **Cocaine intoxication**- Tachycardia, tachypnea, hypertension, headache, chest pain and respiratory depression.
- 5) **Heroin intoxication**- Pinpoint pupils, clammy skin, respiratory depression

88. A bipolar patient was on olanzapine which they discontinued. Relapses with depression what do you do?

A. Restart olanzapine

- B. Start fluoxetine,
 - C. start lamotrigine
- (no quetiapine in the options)

1. Fluoxetine + Olanzapine
2. Quetiapine only
3. Olanzapine only
4. Lamotrigine only / Lurasidone only
5. Lithium only

89. What's definition of standard error of mean?

Glossary of Statistical Terms 3 Identify the correct terminology used in statistics for each of the following definitions given below: The term used to describe a distribution whose values shows either a negative or a positive skew. It is the square root of the variance It is a measure of how close the sample mean is likely to be to the population mean. The term used to describe a distribution in which the values are dispersed equally on either side of the mean.	Choose... Mean Standard error of the mean Parametric Standard deviation Median Non-parametric Quartile Range Mode Interquartile range
---	--

Parametric distribution is a distribution in which the values are distributed equally on either side of the mean. In a parametric distribution the mean = median = mode.

Non-parametric distribution refers to any distribution other than a normal distribution. The distribution of values shows either a negative skew or a positive skew.

Standard deviation and variance are measured of the spread of values around the mean. The standard deviation (S.D.) is the square root of the variance. In a parametric / normal distribution, 68% of values lie within 1 S.D. on either side of the mean. About 95% of values lie within 2. S.D. (1.96 S.D., to be specific) on either side of the mean.

Standard error of the mean refers to the S.D. of a distribution of sample means. It is a measure of how close the sample mean is likely to be to the population mean.

The correct answer is: The term used to describe a distribution whose values shows either a negative or a positive skew. → **Non-parametric**, It is the square root of the variance → **Standard deviation**, It is a measure of how close the sample mean is likely to be to the population mean. → **Standard error of the mean**, The term used to describe a distribution in which the values are dispersed equally on either side of the mean. → **Parametric**

90. number of new cases of dementia is slowly declining. People live longer with the disease. What can be said? **Incidence decreases**

91. New medication reducing mortality rate, but not remitting disease = **Prevalence increases**

92. patient on selegiline aspirin for 3 years but MOCA is low and dementia for 1 year, parkinsons dementia

Dementia- subtypes For each of the following descriptions choose the most appropriate diagnosis A 60-year-old woman has been wandering the streets and was taken to A&E. She appears unconcerned during the assessment. She has diabetes and hypertension. CT shows a subcortical pattern of excessive white matter lesions with no other notable findings. A 70-year-old man with a history of recent onset cognitive decline and visual hallucinations. He has been attending the neurology clinic for the past three years, and his medication includes a list of drugs also to Selegiline. A 67-year-old man with a history of progressive cognitive decline, loss of visuospatial and calculation abilities and reduction in functioning.	Choose... Normal pressure hydrocephalus Depressive pseudodementia Alcoholic dementia Dementia in Pick's disease Wilson's' disease. Dementia in Huntington's disease Lewy body Dementia Binswanger's disease Parkinson's disease dementia Sporadic CJD Alzheimer's disease
--	---

1) Case 1 is **Binswanger's disease** (a progressive small vessel disease). This is subcortical dementia with a clinical course characterized by slow intellectual decline and generalised slowing. There is a history of vascular risk factors like diabetes and hypertension. The clinical picture may be dominated

by dementia, and there may be concomitant physical problems such as gait disorders or dysarthria. MRI scan shows small distinct infarcts (lacunae) or more generalised white matter changes (Leukoaraisis).

2) This patient is on Selegiline, which is a drug used for Parkinson's disease and has been reviewed by the neurology team. It is a case of **Parkinson's disease dementia**. If the parkinsonian symptoms have existed for more than 12 months before dementia develops then a diagnosis of Parkinson's disease dementia is given. Otherwise, you can consider Lewy Body Dementia as an option.

3) It is a case of **Alzheimer's dementia** featured a history of progressive cognitive decline and reduction in functioning. It is a type of cortical dementia involving temporoparietal dysfunction such as loss of visuospatial and calculation abilities and reduction in functioning.

93. Patient presenting with memory loss, dx of high BP MRI shows infarcts- Binswanger?

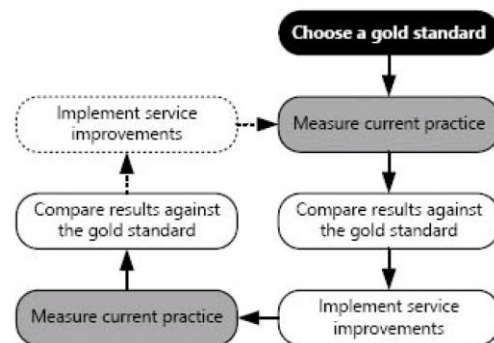
94. One doctor does an audit in a department, presents the results, few months later does a re audit. Whats missing in the loop? **Implementation of change**/ hypothesis generation .

[SPMM] A consultant asks you to complete an audit on a prescribing nicotine replacement therapy for new inpatients. You identify the Trust's Drug & Therapeutics Guideline as the gold standard and collect the data. You then present it to your consultant colleagues. Six months later you reaudit the process to determine if there has been an improvement. What important aspect of the audit process has been omitted?

Select one:

- A. Publishing the results in a peer-reviewed journal
- B. Presenting the findings at a national level
- C. Using NICE guidelines
- D. Developing a hypothesis
- E. Initiating change after the first cycle**

Before re-auditing, one must **implement changes** to see if they have made a difference.



95. 30 year old man, speech difficulties, gait issues, nystagmus, granddad had similar issues and died aged 41 from MI? **Friedreich ataxia**

[SPMM] A 30-year-old man with clumsy gait, repeated falls and visual problems reports that his grandfather reportedly developed the same condition and died at an age of 41 after myocardial infarction. The most likely diagnosis to be considered is

Select one:

- A. Wilsons disease
- B. MELAS syndrome
- C. Fragile X syndrome
- D. Myotonic dystrophy
- E. Friedreich's ataxia**

In Friedreich's ataxia the spinal and peripheral nerves and cerebellum degenerate, affecting balance and movement and sensory functions but with intact cognition. The disorder also causes problems in the heart and spine, and some people with the condition develop diabetes. Various forms of heart disease often accompany Friedreich's ataxia, including hypertrophic cardiomyopathy, myocardial fibrosis, and cardiac failure. Heart block is also common. Heart disease is the most common cause of death.

Friedreich's ataxia is an autosomal recessive, trinucleotide expansion disease affecting GAA sequence of frataxin protein encoding gene on chromosome 9.

96. Mode of given numbers? **39**

Table 1. A stem and leaf plot of the ages of 69 people

[illegible]

Consider the above graph (Can J Psychiatry 1997;42:388–394). From this graph which of the following represents the modal age of the sample? **39**

Consider the above graph (Can J Psychiatry 1997;42:388–394). From this graph which of the following represents the median age of the sample?

To obtain median, first locate the most central observation. As there are totally 69 observations, the most central observation pertains to the $(69/2) + 1 = 35$ th observation. The value of the 35th observation is 37.

97. Infanticide most common in? **Psychosis**

[SPMM] What is the most common type of mental health condition among women who commit infanticide or filicide?

Select one:

- A. Affective disorder**
B. Personality disorder
C. Obsessive-compulsive disorder
D. Schizophrenia
E. Substance use disorder

Neonaticide is the killing of an infant within the first 24 hours of life. Infanticide is used to describe the killing of infants, usually aged under one year. It usually refers to the killing of infants by their mothers. In some jurisdictions, maternal infant killing is subject to special ‘infanticide’ laws. Maternal filicide is defined as child murder by the mother. Mental illness, **especially depression and psychosis**, is often part of the picture. Resnick’s review of the world psychiatric literature on maternal filicide found filicidal mothers to have frequent depression, psychosis, prior mental health treatment, and suicidal thoughts. In cases of filicide committed by a parent, mothers are more likely than fathers to have a history of mental disorder and symptoms at the time of the offence, most often affective disorder.

98. What type of study should be used: Lithium, water and something –

- A. **Ecological/**
- B. Cross sectional the way question was written

99. Case control study to avoid attrition bias

[SPMM] Which of the following is an advantage of case-control design over cohort design?

Select one:

- A. Ability to calculate incidence rates
- B. Can be used to determine survival rates
- C. Absence of recall bias
- D. Can be used to study prodromal course of a disease
- E. **Can explore multiple risk factors for the same disease**

Case-control studies are cheaper and faster, compared with other analytic designs. They are useful for studying diseases with long latent periods. They are useful for studying rare diseases since subjects are selected on the basis of disease status. They allow the investigation of multiple exposures simultaneously

100. To study how money incentives lead to medication compliance RCT with individual randomization: correct

Study designs I

For each of the descriptions below, select the most appropriate **study design** from the list provided

To investigate the relationship between domestic abuse and PTSD, patients referred to a psychiatric outpatient clinic with diagnosed PTSD were enquired about domestic abuse. A group of age and sex matched diabetic patients from a medical clinic were also questioned about domestic abuse in the same manner.

A carefully selected group of parents of children attending Child and Adolescent clinic were invited to provide their views on waiting time for psychotherapy and choices of treatments offered.

A survey was conducted in an unsystematic sample of ward staff at a Mental Health Trust. This survey set out to find the degree of satisfaction with jobs at inpatient psychiatric units

Choose...

- Cohort study
- Randomized controlled trial
- Clinical audit
- Cross-sectional Study
- Case-control study
- Epidemiological study
- Ecological study
- Qualitative study

Q1. In this study, patients with PTSD - the cases - are recruited alongside those with no PTSD - the diabetic patients who act as controls. Hence, this is a **case-control study**.

Q2. A purposive sample is recruited when opinions/themes are sought in a **qualitative study**.

Q3. Surveys are almost always **cross-sectional**.

101. RCT with cluster randomization- GP surgeries are randomly picked and selected a therapy is tried : correct

102. Cross sectional study

103. Child presenting with weight loss and other symptoms of anorexia for the past 6 months after changing schools. Which treatment is needed as first line?

- A. Supportive counselling or
- B. Family therapy or
- C. CBT or
- D. Interpersonal therapy:

For children and young people (0-18) consider: [Psych-Mentor]

- Anorexia-nervosa-focused family therapy (FT-AN)
- If FT-AN is unacceptable, contraindicated or ineffective for children or young people with anorexia nervosa, consider individual CBT-ED or adolescent focused psychotherapy for anorexia nervosa (AFP-AN).
- Do not offer medication as the sole treatment for anorexia nervosa.

[SPMM] Focused family interventions are particularly useful in which of the following subgroups with eating disorders?

Select one:

- A. Late onset anorexia
- B. Adults with co-morbid physical illnesses
- C. Adolescents with anorexia nervosa**
- D. Adolescents with bulimia
- E. Binge eating disorder

Family interventions that directly address the eating disorder are especially useful for children and adolescents with anorexia nervosa

104. 'When I saw him I knew he had bad intentions' which cognitive distortion?

- A. Mind reading**
- B. Emotional reasoning
- C. Dichotomous thinking

Cognitive distortions
Identify the type of cognitive distortion in each of the following examples.

Tom is a perfectionist. When he makes one mistake, he assumes that he has failed everything.

Sam feels overwhelmed by the amount of work he has to do and thinks there is no point in doing any of it.

Sally is shopping when she has some palpitations. She feels very anxious and expects to die of a heart attack.

George sees his girlfriend talking to another man on the street and immediately concludes that she is cheating on him.

Choose...

Catastrophisation

Minimization

All-or-nothing thinking

Personalization

Mind reading

Magnification

Overgeneralisation

Arbitrary inference

Selective abstraction

The correct answer is: Tom is a perfectionist. When he makes one mistake, he assumes that he has failed everything. → **Overgeneralisation**, Sam feels overwhelmed by the amount of work he has to do and thinks there is no point in doing any of it. → **All-or-nothing thinking**, Sally is shopping when she has some palpitations. She feels very anxious and expects to die of a heart attack. → **Catastrophisation**, George sees his girlfriend talking to another man on the street and immediately concludes that she is cheating on him. → **Arbitrary inference**

Negative automatic thoughts (cognitive distortions) include:-

Negative automatic thought	Comment	Example
Dichotomous thinking	The tendency to see things as black and white rather than shades of grey	'If I don't get a perfect score on this test, I'm a complete failure.'

Negative automatic thought	Comment	Example
Personalisation	Incorrectly assuming that things happen due to us. Attributing external events to oneself when there is actually no causal relationship	'My friend cancelled our plans because I must have done something to upset them.'
Overgeneralisation	Coming to a general conclusion based on a single piece of evidence	'I failed my driving test once, so I'll never be able to drive properly.'
Arbitrary inference	Drawing of an unjustified conclusion	'I saw my co-worker talking to our boss, and now I'm sure they were discussing my performance and planning to fire me.'
Selective abstraction (aka filtering)	Concentrating on the negative while ignoring the positives	'Even though I received several compliments on my presentation, I keep focusing on the one negative comment.'
Catastrophising	Expecting disaster from relatively trivial events	'If I don't get this job, my entire career will be ruined, and I'll never find another opportunity.'
Control Fallacies	Believing we are responsible for everything (internal control fallacy) or nothing (external control fallacy)	'My relationship ended because I didn't do enough to make it work. It's all my fault.'
Fallacy of Fairness	Believing that life is fair	'Life should treat me better because I've always been a good person.'
Blaming	Holding others responsible for our distress	'My team's failure is entirely because of my lazy teammate. I did everything perfectly.'
Shoulds	Preconceived rules we believe (often incorrect) which makes us angry when others don't obey them	'Everyone should agree with my opinion, and it's unfair when they don't.'
Magnification	A tendency to exaggerate the importance of negative information or experiences, while trivialising or reducing the significance of positive information or experiences	'I made a small mistake in my presentation, and now I feel like a complete failure.'
Minimisation	An undervaluation of positive attributes	'I won the competition, but it was just luck. I'm not actually talented.'
Emotional Reasoning	Believing what we feel must be true	'I feel like a burden, so I must be a burden to everyone around me.'
Fallacy of Change	Expecting others to change just because it suits us	'My partner should change their behaviour just because it bothers me.'

Negative automatic thought	Comment	Example
Global Labelling	Exaggerating and labelling behaviour (e.g. when you fail at something, saying 'I'm a loser')	'I made a mistake in this project, so I'm a complete idiot.'
Always Being Right	When the need to be right dominates all other needs	'I have to win every argument and prove that I'm always right.'
Heaven's Reward Fallacy	Expecting our sacrifices will pay off	'If I work overtime every day, I'll eventually get a promotion and be rewarded.'
Magical thinking	Incorrectly believing that our actions influences the outcomes	'If I wear my lucky socks, I'll perform better in the exam.'

Selective abstraction: scenario will mention something positive and something negative and the patient/ subject will only focus on the negative (aka filtering).

In **minimisation:** the scenario will only mention something positive and the patient/ subject will down play this

Overgeneralisation (as compared to magnification) is about something that will happen - i failed the test so i will fail in everything in life. In **overgeneralisation-** the end result is like a multiplication of the initial thing (i failed the test - i will fail in everything) so everything is a multiplication of one thing (the exam failure)

In **catastrophisation** the things mentioned are a sequence (palpitation - will have heart attack and die)

In **magnification** is more about how the subject feels - i failed the test, i feel like a failure.

105. Boss praised slide. “He said it only because he corrected the slide. He didnt mean it.”

- A. Minimising
- B. Overgeneralising
- C. **Selective abstraction**
- D. Labeling
- E. Discounting the positives,: yes

106. “I was doing well with the diet but I had a nice meal today so I will be obese”

- A. Dichotomous thinking/
- B. **Overgeneralising**

107. I got an A from this exam but it does not mean I am intelligent”-

- A. **Minimisation**
- B. Selective abstraction
- C. Overgeneralisation

Cognitive distortions

Identify the type of cognitive distortion in each of the following examples.

Tom is a perfectionist. When he makes one mistake, he assumes that he has failed everything.

Sam feels overwhelmed by the amount of work he has to do and thinks there is no point in doing any of it.

Sally is shopping when she has some palpitations. She feels very anxious and expects to die of a heart attack.

George sees his girlfriend talking to another man on the street and immediately concludes that she is cheating on him.

Choose...

- Catastrophisation
- Minimization
- All-or-nothing thinking
- Personalization
- Mind reading
- Magnification
- Overygeneralisation
- Arbitrary inference
- Selective abstraction

The correct answer is: Tom is a perfectionist. When he makes one mistake, he assumes that he has failed everything. → **Overygeneralisation**, Sam feels overwhelmed by the amount of work he has to do and thinks there is no point in doing any of it. → **All-or-nothing thinking**, Sally is shopping when she has some palpitations. She feels very anxious and expects to die of a heart attack. → **Catastrophisation**, George sees his girlfriend talking to another man on the street and immediately concludes that she is cheating on him. → **Arbitrary inference**

108. Which one is useful test when differentiating between mci and dementia?

A. MMSE

B. Clock drawing

C. Collateral info about daily functioning from patient/

D. Collateral info about daily functioning from patient's wife

109. Stats- 2 groups, researcher is making sure there are same amount of patients in control group and treatment group. **Block randomisation**

[SPMM] A researcher wants to make sure that the size of the control and two intervention groups are more or less equal so that statistical deductions can be meaningful. Which of the following methods of randomisation could ensure this?

Select one:

A. Block randomisation

B. Stratification

C. Cluster randomisation

D. Quasi-randomisation

E. Simple randomisation

Block randomization refers to a method where randomization occurs in a set of specified numbers, say blocks of 6 etc., to ensure equal number distribution between two arms. Larger the block size, lesser the ability to predict allocation.

In cluster randomization, the unit of randomization and analysis is various centres or catchment-zones and not individual patients.

Stratification is classifying to subgroups before randomising within each subgroup to ensure impartial allocation of subgroup factors.

Quasi-randomisation refers to 'randomizing' using even/odd numbers of the date of birth, day of the week patient was seen, etc.

110. Stats- 2 groups, researcher is dividing the groups considering the ages of patients. Stratification?yes

111. Stats- 2 groups, researcher is placing the patients in each group. placing the next patient to the group which has less patients. **Minimisation**

[SPMM] In an allocation scheme, the next allocation of a subject depends on the characteristics of those already allocated. This scheme can be called

Select one:

- A. Matching scheme
- B. Blocking scheme
- C. Haphazard randomisation scheme
- D. Minimisation scheme**
- E. Stratification scheme

In **minimisation** schemes, next allocation depends on characteristics of those already allocated. Allocation of each participant aims to ensure a balance of prognostic factors between groups. The disadvantage is that this method is inferior to proper randomisation as it allocation is somewhat exposed and 'controlled' manually.

112. Which R value indicates strong correlation? 0.7, **-1**, 0.5

Q.20 Fresh Q. 2023

113. Patient is on antipsychotics and he is presenting with tremors. Episodic shakes and gets tingling and numbness prior to episodes, no LOC-

A. Dissociative convulsion,

B. EPSE,

C. Complex partial seizure

No LOC point toward simple partial seizure more than complex partial seizure.

Signs & symptoms favouring psychogenic nonepileptic seizures (PNES)	Signs & symptoms favouring epileptic seizures (ES)
Duration longer than 10 minutes	
Vocalisation during or after seizures, complex, with affective content	Vocalisation at the beginning, primitive, has no emotional expression
Ictal crying	
Occurrence from preictal pseudosleep	Occurrence from sleep
Ictal eye closure	
Side-to-side head or body movement (compared with generalised tonic clonic seizures only)	Postictal stertorous breathing (observed in generalised tonic clonic seizures only)
Pelvic thrusting (frontal lobe partial seizures excluded)	
Rapid regaining of alertness and orientation	Post ictal confusion
Fluctuating course	
Asynchronous jerks (frontal lobe partial seizures excluded)	
Memory recall	

Adapted from Xiang et al. 2019



[SPMM] Mr Jenkins is a 20-year-old man who is being reviewed in first seizure clinic. The neurologist suspects he suffers from non-epileptic attacks (or dissociative seizures) based on the history Mr Jenkins provides. Which ONE of the following patient reported seizure characteristics has the best utility in differentiating non-epileptic attacks from epilepsy?

Select one:

- A. Seizures reported during sleep
- B. Stereotyped attacks
- C. Recall for the period of unresponsiveness
- D. Incontinence
- E. Biting the inside of the mouth

Although patients with Non-Epileptic Attacks (NEA) often report seizures in sleep, when they are captured on telemetry they are inevitably preceded by waking, which again can be recorded electrographically. Thus patient report of seizures during sleep are not necessarily helpful as discriminators and need to be examined further. The following are also not definitive discriminators: Biting the inside of the mouth occurs occasionally in NEA and commonly in epilepsy. Incontinence can occasionally occur in NEA and occurs commonly in epilepsy. Stereotyped attacks occur commonly in both NEA and epilepsy. Patients with NEA often have recall for their period of apparent unresponsiveness, whereas those with epileptic seizures do not. Therefore of the above list, this is the best answer.

114. 35 years patient has mild LD and moderate ADHD. Which treatment is suitable for him?

- A. Start atomoxetine,
- B. Start Lisdexamfetamine,**
- C. CBT,
- D. Refer to cardiologists to review first

1.7.11 Offer lisdexamfetamine or methylphenidate as first-line pharmacological treatment for adults with ADHD.

March 2018 – this is an off-label use of lisdexamfetamine for adults with no ADHD symptoms in childhood. See [NICE's information on prescribing medicines](#). Not all preparations of methylphenidate are licensed for treating symptoms of ADHD in adults. **[2018]**

115. Patient has been on Clozapine for the past 3 months but still presenting with auditory hallucinations. What would be the next option?

- A. Start aripiprazole,
- B. Check Clozapine compliance**

[SPMM] A 28-year-old man is taking clozapine for the last eight months but still has some interfering positive symptoms despite good concordance and optimum plasma levels. What is the best-proven strategy to augment this treatment?

Select one:

- A. Fluoxetine augmentation
- B. Risperidone augmentation
- C. Lithium augmentation
- D. Clonazepam augmentation
- E. Amisulpride augmentation**

Clozapine augmentation has become common practice because an inadequate response to clozapine alone is a frequent clinical event. Clozapine augmentation with Amisulpride has some evidence that suggests this approach may be worthwhile. It may also allow a dose reduction in clozapine for those who have a dose-dependent intolerance.

Maudsley It is recommended that all augmentation attempts are carefully monitored and, if no clear benefit is forthcoming, abandoned after 3–6 months. The addition of another drug to clozapine treatment must be expected to worsen overall adverse effect burden and so continued ineffective treatment is not appropriate. In some cases, the addition of an augmenting agent may reduce the severity of some adverse effects (e.g. weight gain, dyslipidaemia – see below) or allow a reduction in clozapine dose. The addition of **aripiprazole** to clozapine may be particularly effective in reversing metabolic effects.^{18,19} Recently published international consensus guidelines recommend (**after optimising plasma levels**) tailoring augmentation agent choice to residual symptoms; adding **amisulpride** or **aripiprazole** for positive symptoms, antidepressants for negative symptoms, and mood stabilisers for suicidal ideation or aggression.

116. What is approved by NICE guidelines for the treatment of refractory OCD?

- A. Anterior cingulotomy,
- B. DBS

Neurosurgery in OCD:

- ➔ **Anterior Cingulotomy** ➔ has been used for OCD ➔ If not in the options then choose DBS
- ➔ rTMS has small evidence and more studies required, has been used in UK for treatment of depression but not OCD

Evidence on the safety and efficacy of deep brain stimulation for chronic, severe, treatment-resistant obsessive-compulsive disorder (OCD) in adults is inadequate in quality and quantity. Therefore, this procedure should only be used in the context of research

117. Comparison of ECT and TMS, which one is correct?

- A. ECT is more effective compared to rTMS**
- B. ECT needs to be given more frequently
- C. ECT targets more specific areas in brain

118. What is the remission rate with ECT in treatment resistant depression? 90% **70%** 50%

It is the most effective short-term treatment for major depression and significantly more effective than sham ECT and antidepressants. **Remission (recovery to previous state of well-being) rates of around 60–80% have been reported when it is used as first-line treatment in a severe depressive episode and remission rates are even higher in psychotic depression.**

Despite its effectiveness in the acute episode, without maintenance treatment the relapse rate is extremely high (**over 80%**) in the 6 months after successful ECT. This can be significantly reduced by either pharmacotherapy, including the combination of nortriptyline and lithium, or continuation ECT and it has been shown that such continuation ECT is not associated with adverse memory outcomes.

119. A patient who tested negative but he actually has the condition. **False negative**

120. What is the likelihood of a patient who tested negative not having the disease: **NPV** (LR was not in the options) - *If LR in the options it should be the answer*

Diagnostic test measures

Lead In: Identify the terms used in diagnostic test measures:

The ability of a test to correctly identify true positives among the ill subjects

The ability of a test to correctly identify true negatives among the healthy subjects

The proportion who are ill among all subjects who are picked up by a test as having the disorder

The proportion who are truly well among all subjects who are declared to be well by a test

This helps to determine how the result of a diagnostic test influences the probability that the patient has the disorder.

Choose...

Positive predictive value
False positivity rate
Total Positives
Specificity
Odds ratio
Negative predictive value
Likelihood ratio
Relative risk
Sensitivity

Sensitivity is the proportion of the 'diseased subjects', who are picked up by the test as having the disorder. In other words, it is the ability of the test to identify those who really have the disorder.

Specificity is the proportion of 'well subjects', who are excluded by the test as not having the disorder. In other words, it is the ability of the test to identify those who really do not have the disorder.

Positive predictive value (PPV) is defined as the proportion of positive test results for which disease is confirmed. In other words, it is the proportion of the test positives who are true positives.

Negative predictive value (NPV) is the proportion of negative test results for which a disease is absent. In other words, it is the proportion of the test negatives who are true negatives.

Likelihood ratio summarises how many times more (or less) likely patients with the disease are to have that particular result than patients without the disease. It is the ratio of the **probability of the specific test result** in people who do have the disease to the probability in people who do not.

[SPMM] The ability of a test to correctly classify subjects as being with or without a disease is called
Select one:

- A. Specificity
- B. Likelihood ratio**
- C. Positive predictive value
- D. Sensitivity
- E. Negative predictive value

This is the likelihood that a test result would be expected in patients with a condition, divided by the likelihood that that same result would be expected in patients without that condition. Imagine a blood test for anorexia, if the LR for a positive test is 5, it means if the test is positive in that patient, then that patient is five times more likely to have anorexia than not have it.

Measure	Formula	Plain english
Sensitivity	$TP / (TP + FN)$	Proportion of patients with the condition who have a positive test result
Specificity	$TN / (TN + FP)$	Proportion of patients without the condition who have a negative test result

Measure	Formula	Plain english
Positive predictive value	$TP / (TP + FP)$	The chance that the patient has the condition if the diagnostic test is positive
Negative predictive value	$TN / (TN + FN)$	The chance that the patient does not have the condition if the diagnostic test is negative
Likelihood ratio for a positive test result	$\text{sensitivity} / (1 - \text{specificity})$	How much the odds of the disease increase when a test is positive
Likelihood ratio for a negative test result	$(1 - \text{sensitivity}) / \text{specificity}$	How much the odds of the disease decrease when a test is negative

121. Child presenting with mild depression, what is the next step?

A. Digital CBT,

B. watchful waiting for 2 months

(watchful waiting should be for 2 weeks as per NICE guidelines, not for 2 months)

Severity of depression	Recommendation
Mild depression (including dysthymia) - without significant comorbid problems or active suicidal ideas or plans	Watchful waiting (no intervention and review in 2 weeks) Digital CBT Group CBT Group IPT Group non-directive supportive therapy (NDST) (If not suitable then individual CBT or attachment-based family therapy)
Moderate to severe depression (continued mild depression despite 2-3 months of psychology is treated as moderate / severe)	5-11 year olds: Family based IPT Family therapy (family-focused treatment for childhood depression and systems integrative family therapy) Psychodynamic psychotherapy Individual CBT +/- fluoxetine 12-18 year olds: Individual CBT +/- fluoxetine (If not suitable then IPT for adolescents, family therapy (attachment-based or systemic), brief psychosocial intervention or psychodynamic psychotherapy)
Treatment resistant depression or depression with psychotic symptoms	Intensive psychological therapy +/- fluoxetine, sertraline, citalopram, augmentation with an antipsychotic

122. Which one is necessary for testamentary capacity?

- A. No mental health disorder, no dementia,
- B. knowing the exact extent and value of their property,**
- C. knowing the legalities,
- D. knowing the beneficiaries?

[Psych-Mentor] Testamentary capacity refers to a person's legal and mental ability to make a will and is part of common law.

There are four requirements for testamentary capacity, at the time of making the will a testator must know:

- The extent and value of their property
- Who the natural beneficiaries are
- The disposition they are making
- The plan as to how the property will be distributed

123. Patient with Parkinson's presenting with low mood, what should be the first step? **Blood tests to exclude an organic cause**

124. A question about body dysmorphic disorder in adolescents cannot remember exactly but options were

- A. It will resolve spontaneously
- B. Increased risk of self-harm and suicide**
- C. No impact on life

125. Which one is a sequelae of head injury?

- A. ASD,
- B. Mood disorders,**
- C. Personality disorders

[SPMM] Traumatic brain injury significantly increases the risk of which of the following?

Select one:

- A. Down syndrome
- B. Borderline personality disorder
- C. Asperger syndrome
- D. Psychosis
- E. Depression**

Explanation: **Depression is the most common psychiatric complication of traumatic brain injury (TBI), with prevalence rates of 30-40%.** The aetiology of post-TBI depression is often difficult to determine because it may be driven by both psychosocial and biological factors. Some have proposed that TBI may cause psychosis, but other findings suggest that postinjury psychotic sequelae are relatively indistinguishable from idiopathic psychotic disorders. **One study suggested that TBI may precipitate the development of schizophrenia** in those with a genetic susceptibility, with a meta-analysis reporting a 60% increase in risk for schizophrenia after TBI. However, most of these findings are based on case reports or population-based studies using medical records to ascertain diagnoses. A 5-year prospective study following moderate to severe TBI found that all post-injury cases had a pre-injury history of

psychosis. Findings of prospective studies have reported relatively low rates of psychotic disorders consistent with population rates.

126. What is a risk factor for autism? Maternal smoking

[SPMM] Autism spectrum disorders are characterised by impairments in social interactions and communication and restricted and repetitive interests. They can also be associated with language delay and intellectual impairment. Which of the following is the greatest risk factor for the development of an autism spectrum disorder?

Select one:

- A. Receiving the MMR vaccine
- B. Low APGAR score**
- C. Maternal antidepressant use during pregnancy
- D. Maternal postnatal depression
- E. Maternal smoking

A low 5-minute APGAR score is associated with later ASD. Other perinatal risk factors for ASD are: abnormal presentation, umbilical-cord complications, foetal distress, birth injury or trauma, multiple birth, maternal hemorrhage, summer birth, low birth weight, small for gestational age, congenital malformation, feeding difficulties, meconium aspiration, neonatal anemia, ABO or Rh incompatibility, and hyperbilirubinemia. Overall, **genetic factors are the greatest known risk factors for ASD**

[SPMM] A child is hyperactive and distractible in school and at home. Choose the risk factor associated with the development of this condition.

Select one:

- A. Maternal benzodiazepine use in pregnancy
- B. Measles vaccination
- C. Maternal SSRI use in pregnancy
- D. Maternal smoking in pregnancy**
- E. Maternal cannabis use in pregnancy

Explanation: ADHD has a heritability of 70-80%, and the risk of ADHD in siblings is 2-3x increased. Rates are increased in premature babies (especially extreme prematurity), babies born to mothers who used substances during pregnancy (e.g. tobacco or alcohol), following head injury, and in some genetic and metabolic disorders. None of the proposed prenatal risk factors have been confirmed as causal for ADHD.

[SPMM] There have been numerous studies into the causes of autism, which have demonstrated that there are several risk factors associated with the condition. Which of the following is the strongest risk factor for an individual developing autism?

Select one:

- A. Small head circumference
- B. Genetic loading**
- C. Socioeconomic status
- D. Younger maternal age

E. Maternal smoking

Genetic loading is the strongest identified risk factor for developing an autism spectrum disorder. Other risk factors include: intellectual impairment; Rett syndrome; Fragile X; immune dysfunction; maternal gestational diabetes; maternal bleeding during pregnancy; maternal medications causing teratogenic effects (especially valproate); antenatal intrauterine infection; foetal presentation; umbilical-cord complications; foetal distress; birth injury or trauma; multiple birth; maternal hemorrhage; summer birth; low birth weight; small for gestational age; low 5 minute Apgar score; meconium aspiration; neonatal anemia; ABO or Rh incompatibility; and hyperbilirubinemia.

National autistic society: Various studies suggest that the ratio of autistic males to females ranges from 2:1 to 16:1. The most-up-to-date estimate is 3:1

NICE: In the UK, the prevalence of ADHD in adults is estimated at 3% to 4%, with a male-to-female ratio of approximately 3:1.

Disorders with higher male preponderance		
Disease	Ratio	Higher in
Lewy body dementia	1.2:1	Males
Paranoid personality disorder	1.3:1	Males
Schizophrenia (Saha et al PLoSMed 2004) incidence rates only. Prevalence is equal.	1.4:1	Males
Parkinson's disease	1.5:1	Males
Conduct Disorder		Males
ADHD	3:1	Males
Alcohol dependence	5:1	Males
Korsakoff's psychosis	2:1	Males
Autism	3:1	Males
Asperger's syndrome Wing 1981 = 10:1	4:1	Males
Antisocial personality disorder	5:1	Males
Tourette's syndrome	4:1	Males

127. What tools to use to assess quality of studies in systematic review? (this was a follow up question on the forest plot of agomelatine, you are asked to review literature) . Options:

- A. **Quorum**,
- B. PRISMA,
- C. CASP
- D. checklist of systematic reviews,
- E. risk of bias.

Reporting guidelines for clinical studies

Choose the most appropriate recommendation to report research studies from the below list:

Reporting of meta-analysis of clinical RCTs	Choose...
Reporting cohort studies	Choose...
Reporting systematic reviews	Choose...

The correct answer is: Reporting of meta-analysis of clinical RCTs → **QUOROM**, Reporting cohort studies → **STROBE**, Reporting systematic reviews → **PRISMA**

Guideline	Topic of guidelines
CONSORT (Consolidated Standards of Reporting Trials)	Randomised controlled trials
TREND (Transparent Reporting of Evaluations with Non-randomized Designs)	Non-randomised controlled trials
PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses)	Systematic Reviews and Meta-Analyses (Replaced QUORUM - Quality of Reporting of Meta-Analyses)
MOOSE (Meta-analysis Of Observational Studies in Epidemiology)	Observational Studies in Epidemiology
STROBE (Strengthening the Reporting of Observational Studies in Epidemiology)	Observational Studies in Epidemiology
SQUIRE (Standards for Quality Improvement Reporting Excellence)	Quality improvement studies
STARD (Standards for the Reporting of Diagnostic accuracy studies)	Diagnostic studies
MIAME (Minimum Information about a Microarray Experiment)	Microarray studies
COREQ (Consolidated criteria for reporting qualitative research)	Qualitative studies

128. Question about ITT in RCT. 2 groups of patients. Some patients were allocated to the antipsychotics group and the other group was placebo. 15 patients initially agreed to take antipsychotics but then they changed their minds before starting to receive treatment. How should you proceed?

A. Include all patients in RCT analysis

B. Exclude dropouts from the final analysis

C. Include dropouts in placebo group

129. What does interpersonal therapy focus on for a person with depression? **Grief and loss**

RIIG

- Role transitions
- Interpersonal deficits
- Interpersonal disputes
- Grief

130. Social anxiety and ASD. Options were:

- A. A diagnosis of social anxiety cannot be made in patients with autism as it is an exclusion criteria
- B. Patients with high functioning autism and aspergers experience more anxiety
- C. Patients with more severe forms of ASD tend to experience more anxiety**
- D. It is nearly impossible to make a diagnosis of social anxiety in the presence of ASD

131. Which one of the following point towards factitious disorder?

- A. Patient does not agree with invasive investigations
- B. Patient has good knowledge of symptomatology of the disease**
- C. Patient has an increased number of visitors at the hospital
- D. Does not give consent to check records

[SPMM] Some patients intentionally falsify their symptoms and past history, and fabricate signs of physical or mental disorder, with the primary aim of obtaining medical attention and treatment. Which of the following diagnoses is most appropriate in these circumstances?

Select one:

- A. Munchhausen syndrome**
- B. Dissociative neurological symptom disorder
- C. Hypochondriasis
- D. Asperger syndrome
- E. Bodily distress disorder

Explanation: In Munchhausen syndrome (**factitious disorder imposed on self in ICD-11**), patients **intentionally falsify their symptoms and past history**, and fabricate signs of physical or mental disorder, with the primary aim of obtaining medical attention and treatment. The diagnostic features are the intentional and conscious production of signs, falsification or exaggeration of the history, and the lack of gain beyond medical attention and treatment.

132. A motivational syndrome- **cannabis**

133. (not sure if this was a question or if I'm confabulating!) can it be about clozapine and continuation of symptoms, I think it was the q with the answer check **clozapine compliance first**

134. First episode psychosis and cardiac problem, which antipsychotic should be given? I think positive symptoms too. = **Aripiprazole**

Effects of antipsychotics on QTc

No effect	Low effect	Moderate effect	High effect
- Brexipiprazole - Cariprazine - Lurasidone - Lumateperone	- Aripiprazole - Asenapine - Clozapine - Flupentixol - Fluphenazine - Perphenazine - Prochlorperazine - Olanzapine - Paliperidone - Risperidone - Sulpiride	- Amisulpride - Chlorpromazine - Haloperidol - Iloperidine - Levomepromazine - Melperone - Pimavanserin - Quetiapine - Ziprasidone	- Any intravenous antipsychotic - Pimozide - Sertindole - Any drug or combination of drugs in doses exceeding recommended maximum

Adapted from Maudsley Prescribing Guidelines, 2021. p. 143.

135. Best universal prevention against alcohol dependence? Education in schools, **increase price**

136. Benefit cost ratio question, women had a value of 0.3 and men 17.9. Options:

- A. Women had higher benefit
- B. Men had higher benefit
- C. Men higher benefit

137. Poor Prognostic factors for FEP : Insidious onset, Longer DUP

Prognostic factors in Schizophrenia (with respect to relapses and rehospitalisations)	
Good Prognosis Late onset Obvious precipitating factors Acute onset Good premorbid adjustment Affective symptoms (esp. depression) Being married Family history of affective disorders Good social support Positive symptoms only Good initial response to treatment (best predictor)	Poor Prognosis Early onset* (not strong) No precipitating factors** Insidious onset** Poor premorbid adjustment Social withdrawal Single, divorced, or widowed Family history of schizophrenia* (weak, controversial) Poor social network, High EE families Negative symptoms Poor compliance Neurological signs and symptoms History of perinatal trauma No remissions in 3 years Many relapses History of violence
* Suvisaari, JM. Br J Psychiatry. 1998 Dec;173:494-500. ** Rosen, K & Garety, P. Schizophr Bull . 2005; 31:735-750.	

Adapted from Kaplan and Sadock's Synopsis of Psychiatry, 9th ed. New York, NY: Lippincott Williams & Wilkins; 2003.

[SPMM] Miss Thomson is a 24-year-old woman who has been admitted to an inpatient psychiatry unit after beginning to hear voices coming from her light bulbs and having persecutory beliefs about her housemates. Her parents have attended the ward round and are asking what her likely prognosis will be. Which of the following would be the **worse** prognostic factor in her case?

Select one:

- A. Insidious onset
- B. Long Duration of Untreated Psychosis**
- C. Immigrant
- D. Female gender
- E. Poor compliance with medication

Features that predict worse outcome for FEP patients include:

- Longer duration of untreated psychosis (DUP).
- Male gender,
- Nonadherence, poor premorbid adjustment.
- The insidious mode of onset,
- The greater severity of negative symptoms,
- Comorbid substance use disorders and
- Having non-affective psychosis

138. Recovery model- assessing patient's strengths

[SPMM] Your next work placement will be with the Recovery Team. Which of the following is a guiding principle of the recovery model?

Select one:

- A. Ascertaining problems the patient is experiencing
- B. Undertaking regular psychiatric assessments
- C. Ensuring the patient attends their appointments
- D. Checking adherence to treatment
- E. **Assessing the patient's strengths**

Recovery teams are community mental health teams for people aged 18-65 years with severe and enduring mental illness, who have complex needs. Aims of the team are to help patients function independently and as well as possible in everyday life within the community. The team uses the recovery model, which emphasizes and supports a person's potential for recovery. The model is strengths-based, which means the focus is on the strengths and resources of the patient rather than their impairment.